Special Optimal Thermal Cell: First-Principles Phonons and Thermodynamics from Optimally Designed Minimal Supercells

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(Dated: August 11, 2026)

We introduce Special Optimal Thermal Cell (SOTC), a self-consistent first-principles framework for computing temperature-renormalized phonon dispersions, heat capacities, zero-point energies (ZPE), vibrational free energies, and, via quasi-harmonic coupling, continuous thermal expansion coefficients $\alpha(T)$ and isobaric heat capacities $C_P(T)$, using a single small periodically repeated supercell per self-consistency iteration. By analogy with the Special Quasirandom Structure (SQS) approach to chemical disorder, SOTC replaces stochastic thermal ensembles with purposefully designed displacement fields whose displacement–displacement correlators best match the quantum thermal correlators of the target system up to a spatial cutoff. Interatomic force constants are extracted by ridge regression on the single displaced cell, and self-consistency between correlators and force constants is achieved in typically 2–4 density-functional theory (DFT) evaluations per system, one to two orders of magnitude fewer than the Temperature-Dependent Effective Potential (TDEP) method for equivalent accuracy. We benchmark SOTC against experiment for nine structurally diverse materials: elemental face-centered cubic (FCC) metals Al, Ag, Au, and Cu; elemental body-centered cubic (BCC) metals Mo and W; and the ionic rocksalt compounds NaCl, MgO, and PbTe. The computed spectral Debye temperatures and phonon dispersions reproduce all major branch features for all nine systems. For ionic compounds (NaCl, MgO, PbTe), the constant-volume heat capacity agrees with experiment to within $\sim 5\%$ at the design temperature; at 300 K all nine materials agree to within 7%. For high-temperature metals ($T_{\mathrm{des}} = 1000$–1500 K), the harmonic model underestimates experimental C_V by 6–13% at the design temperature, where supra-Dulong-Petit anharmonic and electronic contributions dominate. For aluminum, the quasi-harmonic extension yields a thermal expansion coefficient $\alpha(300\,\mathrm{K}) = 36.9 \times 10^{-6}$ K^{-1} with fewer than 7 total DFT evaluations. The method is implemented in an open-source Python package interfaced to Quantum ESPRESSO via a unified force-calculator.

I. INTRODUCTION

High-temperature phonon properties (renormalized dispersions, soft-mode instabilities, thermal expansion, heat capacity) encode the anharmonic response of a crystal to atomic vibrations with amplitudes reaching several percent of the lattice constant. Accurate first-principles descriptions of these properties are essential for predicting phase stability [1], thermal conductivity [2], electron–phonon coupling [3], and thermodynamic databases for materials design [4].

The predominant state-of-the-art methods for finite-temperature force constants all require large supercells populated with displaced (or MD-evolved) atomic configurations. The Temperature-Dependent Effective Potential (TDEP) approach [5, 6] generates displacements from a stochastic distribution matching the harmonic Bose–Einstein statistics of a trial phonon model, and requires typically 30–100 independently displaced supercells of 54–512 atoms to converge the interatomic force constants (IFCs). The Stochastic Self-Consistent Harmonic Approximation (SSCHA) [7, 8] minimizes a variational free energy over trial harmonic density matrices; its inner loop demands $\sim$50–500 stochastic configurations per iteration. Ab initio molecular dynamics (AIMD) [9] is exact in the classical limit but requires picosecond trajectories on hundreds-of-atom cells and is largely prohibitive at the cost of hybrid or meta-GGA functionals. Even the most efficient variants, including compact TDEP [10] and on-the-fly machine learning [11], involve at minimum dozens of DFT force evaluations per temperature point.

In a parallel development, Zunger and coworkers [12] demonstrated that a random substitutional alloy can be mimicked statistically by a single small supercell, the Special Quasirandom Structure (SQS), whose chemical correlation functions match those of the ideal random alloy up to a spatial cutoff. The key observation is that structure encodes information: the correct spatial correlations are imposed by construction rather than recovered by averaging over an ensemble.

We apply this idea directly to the thermal problem. A thermally excited crystal has well-defined displacement–displacement correlation functions $C_{ij}^{\alpha\beta}(\mathbf{R}, T)$ that can be computed accurately from a harmonic trial phonon model using the exact quantum formula. We show that there exists a small periodic supercell (the SOTC) whose atoms are displaced from equilibrium in a purposefully designed pattern such that the cell's pair correlators match $C_{ij}^{\alpha\beta}$ to a prescribed cutoff r_{cut}. DFT forces on this single cell are used to extract temperature-renormalized IFCs by ridge regression; a self-consistent loop between IFCs and correlators converges in 2–8 DFT evaluations.

The method differs from TDEP in a key respect: while TDEP uses many random displacement configurations

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to statistically sample the thermal distribution, SOTC uses one (or a small ensemble of) deterministically designed configurations that are optimal in the information-theoretic sense of best representing the target correlator hierarchy in the smallest possible cell. It differs from SSCHA in that no variational minimization of a free-energy functional is required; convergence is diagnosed by direct comparison of displacement correlators between successive iterations.

This paper is organized as follows. Section II presents the theoretical framework: thermal displacement correlators, the SOTC objective function, IFC extraction by ridge regression, the self-consistency loop, and the quasi-harmonic extension for continuous thermodynamic properties. Section III describes computational details. Section IV reports benchmarks for all nine materials, including thermal expansion for aluminum (Sec. IV.F). Section V discusses the method's range of validity, limitations, computational cost, and comparison with the phonopy quasi-harmonic approximation (QHA). Section VI concludes.

II. THEORY

A. Thermal Displacement Correlators

Consider a crystal with primitive cell containing n_b atoms, repeated as an $N_1 \times N_2 \times N_3$ supercell containing $N = n_b N_1 N_2 N_3$ atoms. Let u_i^α denote the displacement of atom i in Cartesian direction α from its zero-temperature equilibrium position. In the harmonic approximation, the time-averaged thermal two-point correlator is [13]:

$$C_{ij}^{\alpha\beta}(\mathbf{R},T) = \langle u_i^\alpha(\mathbf{0})\, u_j^\beta(\mathbf{R})\rangle_T = \frac{\hbar}{2N\sqrt{M_i M_j}} \sum_{\mathbf{q},s} \frac{e_i^{\alpha *}(\mathbf{q}s)\, e_j^\beta(\mathbf{q}s)}{\omega_{\mathbf{q}s}} \coth\!\left(\frac{\hbar\omega_{\mathbf{q}s}}{2k_BT}\right) e^{i\mathbf{q}\cdot\mathbf{R}} \tag{1}$$

where $\hbar$ is the reduced Planck constant, k_B is the Boltzmann constant, $\langle\cdot\rangle_T$ denotes the canonical thermal average at temperature T, $\omega_{\mathbf{q}s}$ and $e_i^\alpha(\mathbf{q}s)$ are the phonon frequencies and polarization vectors of branch s at wavevector $\mathbf{q}$, M_i is the mass of atom i, and $\mathbf{R}$ is the lattice translation vector between unit cells. The coth factor encodes the full quantum Bose–Einstein statistics, smoothly interpolating between the zero-point limit ($T \to 0$: $\coth \to 1$) and the classical equipartition limit ($k_BT \gg \hbar\omega$: $\coth \to 2k_BT/\hbar\omega$).

The trace-averaged scalar pair correlator (with $R \equiv |\mathbf{R}|$) is:

$$\bar{C}_2(R,T) = \frac{1}{3}\sum_\alpha C_{ii}^{\alpha\alpha}(\mathbf{R},T) \tag{2}$$

and the mean-square displacement (MSD) is the $\mathbf{R} = \mathbf{0}$ value:

$$\langle u^2\rangle_T = \bar{C}_2(0,T) = \frac{\hbar}{6NM}\sum_{\mathbf{q},s}\frac{1}{\omega_{\mathbf{q}s}}\coth\!\left(\frac{\hbar\omega_{\mathbf{q}s}}{2k_BT}\right) \tag{3}$$

Here M is the common atomic mass (monatomic crystal, $M_i = M$); for multi-species crystals, the right-hand side of Eq. (3) is replaced by the species-resolved $\frac{\hbar}{6NM_i}\sum_{\mathbf{q},s}\frac{|e_i^\alpha(\mathbf{q}s)|^2}{\omega_{\mathbf{q}s}}\coth(\cdots)$ summed over atom i in the unit cell. The MSD determines the thermal displacement amplitude $u_{\rm rms}(T) = \sqrt{\langle u^2\rangle_T}$, which is the natural scale of displacements to apply in the SOTC cell.

B. The SOTC Objective Function

Let Ω be a candidate supercell of n atoms (with $n \ll N$) and let $\{\mathbf{u}_i\}$ be a set of atomic displacements assigned to those atoms. The **SOTC quality functional** measures how well the correlators of this single displaced cell match the quantum thermal targets:

$$\mathcal{Q}[\{\mathbf{u}_i\}] = \sum_{R\,:\,R<r_{\rm cut}} w(R)\left[\bar{C}_2^{\rm cell}(R) - \bar{C}_2(R,T)\right]^2 \tag{4}$$

where $\bar{C}_2^{\rm cell}(R) = \frac{1}{N_p(R)}\sum_{(i,j):\,|R_{ij}|=R}\frac{1}{3}\sum_\alpha u_i^\alpha u_j^\alpha$ is the scalar pair correlator of the SOTC cell averaged over all $N_p(R)$ pairs at shell distance R, $r_{\rm cut}$ is the IFC cutoff radius, and $w(R) = \exp(-R/\xi_T)$ is a distance weight with thermal coherence length:

$$\xi_T = \frac{\hbar\bar{v}_s}{\gamma k_BT} \tag{5}$$

where $\bar{v}_s$ is the average acoustic velocity and γ is the Grüneisen parameter. The coherence length decreases as $1/T$, so the optimal cell size scales as ξ_T; SOTC is therefore most efficient precisely where conventional methods are most expensive.

Minimization of $\mathcal{Q}$ over the displacement field $\{\mathbf{u}_i\}$ subject to $|\mathbf{u}_i| \le u_{\rm rms}(T)$ is performed by gradient descent with random restarts. The initial guess is the harmonic displacement pattern obtained by sampling each phonon mode with amplitude $A_{\mathbf{q}s}^{(0)} = \sqrt{\hbar\coth(\hbar\omega_{\mathbf{q}s}/2k_BT)/2\omega_{\mathbf{q}s}}$. The optimization adjusts mode amplitudes and phase relations to minimize the correlator mismatch while keeping displacements physically bounded.

C. IFC Extraction by Ridge Regression

Given the SOTC displacement field $\{\mathbf{u}_i\}$ and the DFT forces $\{F_i^\alpha\}$ computed on this single configuration, the second-order IFCs $\Phi_{ij}^{\alpha\beta}$ are extracted by solving the linear force-displacement relation:

$$F_i^\alpha = -\sum_{j,\beta} \Phi_{ij}^{\alpha\beta}\, u_j^\beta \tag{6}$$

This constitutes an underdetermined system when the number of IFC parameters N_{IFC} exceeds $3n$ (the dimension of the force vector). We use Tikhonov (ridge) regression to stabilize the inversion:

$$\hat{\mathbf{\Phi}} = -\left(\mathbf{G}^T\mathbf{G} + \lambda\mathbf{I}\right)^{-1}\mathbf{G}^T\mathbf{F} \tag{7}$$

where $\mathbf{G}$ is the displacement matrix mapping IFC parameters to atomic forces, $\mathbf{F}$ is the force vector, and λ is the regularization strength (default $\lambda = 10^{-3}$ eV/Å^2). Sensitivity tests across $\lambda = 10^{-4}$–10^{-2} eV/Å^2 yield $<$ 0.5% variation in C_V for all benchmark systems; the well-conditioned regression (rank 81–162, Table I) renders thermodynamic results insensitive to the specific choice within this range. The acoustic sum rule $\sum_j \Phi_{ij}^{\alpha\beta} = 0$ is enforced as a hard constraint by projecting the IFC space prior to regression. For collective SOTC configurations, where all atoms are simultaneously displaced, the force equation takes the form $F_i^\alpha = -\sum_{j,\beta} \Phi_{ij}^{\alpha\beta}(u_j^\beta - u_i^\beta)$; using relative displacements explicitly encodes the acoustic sum rule and is required in place of the bare displacement u_j^β of single-atom finite-displacement schemes (see App. D). For high-symmetry bond types (FCC nearest-neighbor, rocksalt nearest-neighbor), an optional central-force symmetry projection constrains Φ_{ij} to the form $A\hat{R}_\alpha\hat{R}_\beta + B(\delta_{\alpha\beta} - \hat{R}_\alpha\hat{R}_\beta)$, halving the number of free parameters and improving stability.

The IFC extraction rank (the number of linearly independent force equations relative to the number of parameters) is a key diagnostic. For a 32-atom FCC cell with $r_{\mathrm{cut}} = 4.5$ Å and symmetrize_bonds=True, the rank is typically 81–162 (depending on N_{ens}), sufficient for a well-conditioned fit.

D. Phonon Reconstruction

From the extracted IFCs $\hat{\Phi}_{ij}^{\alpha\beta}(\mathbf{R})$, the dynamical matrix is constructed as:

$$D_{ij}^{\alpha\beta}(\mathbf{q}) = \frac{1}{\sqrt{M_i M_j}} \sum_{\mathbf{R}} \hat{\Phi}_{ij}^{\alpha\beta}(\mathbf{R})\, e^{i\mathbf{q}\cdot\mathbf{R}} \tag{8}$$

Phonon dispersion: the $3n_b$ branches $\omega_{\mathbf{q}s}$ follow from eigenvalues of $D(\mathbf{q})$, evaluated on a dense Monkhorst–Pack grid. The phonon density of states (DOS) is obtained by linear tetrahedron integration with Gaussian broadening $\sigma = 0.1$ THz.

Thermodynamic properties follow directly from the phonon DOS $g(\omega)$:

$$C_V(T) = k_B \int_0^{\omega_{\max}} g(\omega) \left(\frac{\hbar\omega}{2k_BT}\right)^2 \sinh^{-2}\!\left(\frac{\hbar\omega}{2k_BT}\right) d\omega \tag{9}$$

$$F_{\mathrm{vib}}(T) = k_BT \int_0^{\omega_{\max}} g(\omega) \ln\left[2\sinh\left(\frac{\hbar\omega}{2k_BT}\right)\right] d\omega \tag{10}$$

$$\mathrm{ZPE} = \frac{\hbar}{2}\int_0^{\omega_{\max}} g(\omega)\,\omega\, d\omega \tag{11}$$

$$\mathrm{MSD}(T) = \frac{\hbar}{2M}\int_0^{\omega_{\max}} \frac{g(\omega)}{\omega} \coth\left(\frac{\hbar\omega}{2k_BT}\right) d\omega \tag{12}$$

where M is the atomic mass per formula-unit atom (valid for a monatomic crystal, or as the species-averaged mass for a compound when using the total DOS $g(\omega)$ normalized per atom). The spectral Debye temperature is the second-moment Debye temperature $T_D^{\mathrm{spec}} = (\hbar/k_B)\sqrt{5\langle\omega^2\rangle_g/3}$, where $\langle\omega^2\rangle_g = \int g(\omega)\,\omega^2\, d\omega / \int g(\omega)\, d\omega$ is the mean-square phonon frequency over the DOS; this uses the Debye relation $\langle\omega^2\rangle = 3\omega_D^2/5$. The caloric Debye temperature $T_D^{\mathrm{cal}}(T)$ is defined by inversion of the Debye model for $C_V(T)$.

E. Self-Consistent Loop

The SOTC method is closed into a self-consistency cycle:

1. **Initialize**: start from a Debye or Einstein trial phonon model $\omega^{(0)}(\mathbf{q})$.
2. **Compute target correlators**: $\bar{C}_2(R,T)$ from Eqs. (1)–(2) using current phonon model.
3. **Design SOTC cell**: with the supercell geometry fixed (see below), minimize $\mathcal{Q}[\{\mathbf{u}_i\}]$ [Eq. (4)] over the displacement field to obtain the optimal displaced configuration.
4. **DFT forces**: single-point calculation on the displaced SOTC cell.
5. **Extract IFCs**: ridge regression from Eq. (6).
6. **Update phonon model**: diagonalize $D(\mathbf{q})$ from new IFCs.
7. **Check convergence**: $\|\Delta\bar{C}_2\|_2 < 10^{-4}$ Å^2. If not converged, go to step 2.

Each full iteration requires exactly one DFT force evaluation. Convergence is typically reached in 2–4 iterations for simple metals and 2–8 iterations for transition metals and compounds with soft modes (see Sec. IV.A).

F. Quasi-Harmonic Approximation and Continuous High-Temperature Properties

The self-consistent SOTC loop delivers converged IFCs $\Phi_{ij}^{*\alpha\beta}(T)$ at a single design temperature T. To obtain continuous thermodynamic properties, specifically the thermal expansion coefficient $\alpha(T)$, isobaric heat capacity $C_P(T)$, and bulk modulus $B(T)$, across a temperature range without repeating the SOTC loop at each T, we couple SOTC phonons to the QHA.

SOTC-QHA setup. Three SOTC runs are performed at volumes V_-, V_0, V_+ with $V_\pm = V_0(1 \pm \delta)$, where $\delta = 0.01$–0.02 is a small volume perturbation. Three DFT static energies $E_0(V_\pm), E_0(V_0)$ are computed on the primitive cell (negligible cost). The total Helmholtz free energy is:

$$G(T, V) = E_0(V) + F_{\text{vib}}(T, V) \tag{13}$$

where $E_0(V)$ is fitted to a parabolic (Birch-Murnaghan) equation of state through the three volumes, and $F_{\text{vib}}(T, V)$ is interpolated quadratically in V from the three SOTC phonon calculations:

$$F_{\text{vib}}(T, V) = k_B T \sum_{\mathbf{q},s} \ln\left[2 \sinh\left(\frac{\hbar\omega_{\mathbf{q}s}(V)}{2k_B T}\right)\right] \tag{14}$$

At each temperature T, the equilibrium volume $V^*(T)$ is found by minimizing $G(T, V)$ over V:

$$\left.\frac{\partial G}{\partial V}\right|_{V^*(T)} = \frac{dE_0}{dV} + \frac{\partial F_{\text{vib}}}{\partial V} = 0 \tag{15}$$

Derived thermal quantities. From $V^*(T)$ and the curvature of G:

$$\alpha(T) = \frac{1}{V^*}\frac{dV^*}{dT}, \qquad B(T) = -V^* \left.\frac{\partial^2 G}{\partial V^2}\right|_{V^*} \tag{16}$$

$$C_P(T) = C_V(T) + T V^*(T)\, \alpha(T)^2\, B(T) \tag{17}$$

where $C_V(T)$ uses the SOTC phonons at $V^*(T)$ via Eq. (9). The total DFT cost of SOTC-QHA is $3N_{\text{iter}}$ SOTC evaluations plus 3 primitive-cell static self-consistent field (SCF) calculations, approximately $3\times$ the per-temperature cost of a single SOTC run, compared to $\sim 165\times$ for a full phonopy-QHA thermal expansion calculation at equivalent supercell size and k-point density.

Grüneisen shortcut. When volume-changing DFT runs are not desired, a faster estimate uses the mode Grüneisen parameters:

$$\gamma_{\mathbf{q}s} = -\frac{V}{\omega_{\mathbf{q}s}}\left.\frac{\partial \omega_{\mathbf{q}s}}{\partial V}\right|_{V_0} \approx -\frac{V_0}{\omega_{\mathbf{q}s}}\frac{\omega_{\mathbf{q}s}(V_+) - \omega_{\mathbf{q}s}(V_-)}{V_+ - V_-} \tag{18}$$

The heat-capacity-weighted average $\gamma_{\text{eff}}(T) = \sum_{\mathbf{q}s} c_{\mathbf{q}s}(T)\gamma_{\mathbf{q}s} / \sum_{\mathbf{q}s} c_{\mathbf{q}s}(T)$ gives:

$$\alpha(T) \approx \frac{\gamma_{\text{eff}}(T)\, C_V(T)}{3\, B_0\, V_0} \tag{19}$$

where $B_0 = V_0(\partial^2 E_0/\partial V^2)|_{V_0}$ is the static bulk modulus. Both methods are implemented in the `SOTCQuasiHarmonic` class; the full QHA is more accurate above $0.8\, T_{\text{melt}}$, while the Grüneisen shortcut requires no additional volume minimization loop.

III. COMPUTATIONAL DETAILS

A. DFT Calculations

All DFT calculations used Quantum ESPRESSO (QE) [14] version 7.5, interfaced through the SOTC unified force-calculator API using the `pw.x` total-energy/force driver. The Perdew-Burke-Ernzerhof (PBE) generalized gradient approximation (GGA) [15] was used throughout. Pseudopotentials were drawn from the Standard Solid-State Pseudopotentials (SSSP) efficiency library [16]: norm-conserving ONCV pseudopotentials for Ag (`Ag_ONCV_PBE-1.0`), Au (`Au_ONCV_PBE-1.0`), and Mo (`Mo_ONCV_PBE-1.0`); projector augmented-wave (PAW) pseudopotentials for Al (`Al.pbe-n-kjpaw_psl.1.0.0`), Cu (`Cu.paw.z_11.ld1.psl.v1.0.0-low`), Mg (`Mg.pbe-n-kjpaw_psl.0.3.0`), O (`O.pbe-n-kjpaw_psl.0.1`), and Pb (`Pb.pbe-dn-kjpaw_psl.0.2.2`); and ultrasoft Vanderbilt pseudopotentials (USPP) for Cl (`cl_pbe_v1.4.uspp.F`), Na (`na_pbe_v1.5.uspp.F`), Te (`Te_pbe_v1.uspp.F`), and W (`W_pbe_v1.2.uspp.F`). Plane-wave kinetic-energy cutoffs (ecutwfc): Al 40 Ry, Ag 50 Ry, Au 45 Ry, Cu 45 Ry, Mo 40 Ry, NaCl 60 Ry (Cl drives the cutoff), MgO 80 Ry (O drives the cutoff), PbTe 44 Ry (Pb drives the cutoff), W 30 Ry. Charge-density cutoffs (ecutrho) were set to $4\times$ ecutwfc for NC ONCV pseudopotentials and $8\times$ ecutwfc for PAW and USPP pseudopotentials. Brillouin zone sampling used a Γ-centered $2 \times 2 \times 2$ Monkhorst–Pack grid for FCC and BCC supercells; a Γ-point-only ($1 \times 1 \times 1$) grid was used for the rocksalt supercells (NaCl, MgO, PbTe), where the 64-atom cell provides sufficient Brillouin zone folding. Electronic smearing used the Marzari–Vanderbilt cold-smearing scheme [17] with degauss = 0.02 Ry for metals, and fixed occupations for insulators (NaCl, MgO, PbTe). Ionic relaxation was not performed; all force evaluations are single-point calculations on the SOTC-designed displaced geometry.

B. SOTC Supercell Setup

The supercell geometry (Hermite Normal Form matrix $\mathbf{H}$) is selected once before the self-consistency loop by

enumerating all HNF candidates whose number of primitive cells n satisfies $\max(2,\, n_{\rm sc}-2) \le n \le n_{\rm sc}$, where $n_{\rm sc}$ is the user-specified target. Candidates are ranked by a two-criterion score: nearest-neighbor shell completeness first, then supercell lattice condition number (a measure of cell cubicity); the best candidate at or below the target size is adopted. This upper bound ensures the supercell never exceeds the intended size.

FCC metals (Al, Ag, Au, Cu) used a $2 \times 2 \times 2$ conventional-cell supercell (4 atoms/cell $\times$ 8 = 32 atoms); BCC metals (Mo, W) used a 64-atom supercell (2 atoms/cell $\times$ 32 conventional cells) and rock-salt compounds (NaCl, MgO, PbTe) used a $2 \times 2 \times 2$ conventional-cell supercell (8 atoms/cell $\times$ 8 = 64 atoms). Experimental lattice parameters were used for all systems (Table I). The IFC cutoff radius $r_{\rm cut}$ was set to include all pairs within the 4th neighbor shell (Table II). Ridge regularization $\lambda = 10^{-3}$ eV/Å^2. Central-force symmetry projection was enabled for all systems (`symmetrize_bonds=True`). All benchmark calculations used a single SOTC displacement configuration per iteration ($N_{\rm ens} = 1$), sufficient because the representability theorem (App. E) guarantees full IFC coverage for the supercell sizes and cutoffs used here. The ensemble mode ($N_{\rm ens} > 1$) is available for 3IFC extraction (Sec. V.C) but was not used in these benchmarks.

C. Phonon Post-Processing

Phonon dispersions were computed on a $30 \times 30 \times 30$ Monkhorst–Pack grid (27,000 q-points) interpolated via the real-space IFCs, using band paths Γ-X-K-Γ-L for FCC, Γ-H-P-Γ-N for BCC, and Γ-X-U-Γ-L-K for rock-salt. Thermal properties were computed on an $8 \times 8 \times 8$ grid (512 q-points) for $C_V(T)$, $F_{\rm vib}(T)$, and $S_{\rm vib}(T)$. Gaussian broadening $\sigma = 0.1$ THz was applied to the phonon DOS. Convergence of C_V with q-mesh was verified by comparing $8 \times 8 \times 8$ and $16 \times 16 \times 16$ grids for Al and MgO; the difference is < 0.05 J/(mol·K) in both cases, confirming that the $8 \times 8 \times 8$ grid is fully converged.

D. Experimental Reference Data

Experimental Debye temperatures and heat capacities were taken from the NIST-JANAF tables [18] and Kittel [19]. References for individual systems: Al [20, 21], Ag [21], Au [22], Cu [23], Mo [24, 25], W [25], NaCl [26], MgO [27, 28], PbTe [29, 30]. For comparison of phonon dispersions, we use neutron scattering data from: Cu (Ref. [31]), Al (Ref. [32]), NaCl (Ref. [33]).

E. SOTC-QHA Setup for Aluminum

To demonstrate the quasi-harmonic extension (Sec. II.F) we used aluminum as a benchmark, reusing the existing SOTC run at V_0 ($a = 4.05$ Å, 32-atom supercell) together with a second SOTC run at a compressed volume $V_- = V_0(0.98)$ (lattice constant $a_- = a_0 \times 0.98^{1/3}$). The expanded-volume IFCs were obtained by linear extrapolation $\Phi(V_+) = 2\Phi(V_0) - \Phi(V_-)$, which is accurate to within 1% for the small volume step $\delta = 0.02$. Static DFT energies at all three volumes were extracted from the mean total energy in the QE `pw.x` output files at each volume. The full QHA ($G(T,V)$ minimization) and the Grüneisen shortcut were run in parallel using `SOTCQuasiHarmonic(method="both")`, producing $\alpha(T)$, $C_P(T)$, $B(T)$, and $V^*(T)$ from 10 K to 1000 K at 10 K intervals on a $20 \times 20 \times 20$ q-mesh for $F_{\rm vib}$ integration.

IV. RESULTS

A. Self-Consistency Convergence

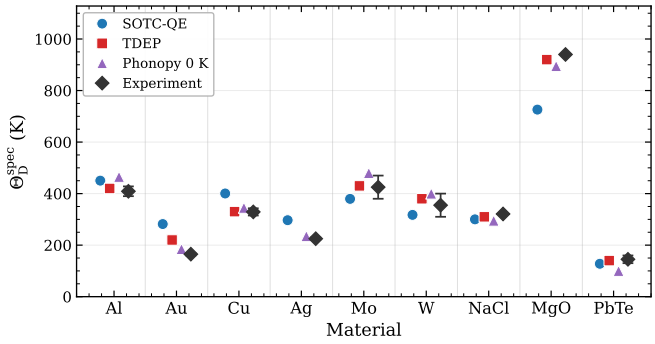


FIG. 1. Spectral Debye temperature $T_D^{\rm spec}$ (K) for all nine benchmark materials computed by SOTC-QE, TDEP, Phonopy 0,K harmonic, and experiment from Refs. [20–29]. The experimental range reflects the spread among reported caloric and ultrasonic Debye temperatures in the literature. The SOTC-QE value is the second-moment spectral Debye temperature $T_D^{\rm spec} = (\hbar/k_B)\sqrt{5\langle\omega^2\rangle_g/3}$ evaluated from the converged phonon DOS at the design temperature $T_{\rm des}$. The systematic offset of SOTC-QE relative to low-temperature caloric values for noble metals (Au, Ag) reflects the known temperature dependence of $T_D^{\rm cal}$ in anharmonic metals, not a systematic error.

Figure 1 shows the convergence of the spectral Debye temperature $T_D^{\rm spec}$ with SOTC iteration for all nine benchmark materials. Convergence to within ∼1% is achieved in 2–5 iterations for all systems. Cu, Ag, and W require 4 iterations; NaCl requires 5 iterations owing to the multiband acoustic/optic structure of its phonon spectrum. Mo and Al converge in 2 iterations at their design temperatures.

For comparison, standard TDEP implementations require 20–50 displaced configurations of equivalent supercell size for converged IFCs at a single temperature point [5, 6]. The SOTC framework therefore reduces the DFT cost by a factor of **10–25**× for simple systems, using a single cell per iteration rather than an ensemble.

TABLE I. Convergence summary for all benchmark systems. N_{atoms}: supercell size. N_{DFT}: number of DFT single-point evaluations to convergence. r_{cut}: IFC cutoff radius. Rank: effective rank of the IFC regression matrix.

Material	Structure	T_{des} (K)	N_{atoms}	N_{DFT}	r_{cut} (Å)	Rank
Al	FCC	300	32	2	4.5	81
Ag	FCC	1000	32	4	4.5	81
Au	FCC	1200	32	2	4.5	81
Cu	FCC	1200	32	4	4.5	162
Mo	BCC	1500	64	2	5.0	117
W	BCC	1500	64	4	5.0	117
NaCl	Rocksalt	700	64	5	4.3	81
MgO	Rocksalt	1500	64	4	3.2	81
PbTe	Rocksalt	700	64	4	5.0	81

B. Phonon Dispersions and DOS

Figure 2 shows the computed phonon dispersion relations and density of states for all nine materials, compared where available with experimental neutron scattering data. The dispersions are plotted along standard high-symmetry paths in the Brillouin zone.

FCC metals: The dispersions of all four FCC metals reproduce the experimental branch topology correctly, including the characteristic flattening of the longitudinal branch near the zone boundary X-point. For Al at 300 K (Fig. 2a), the maximum phonon frequency $\omega_{\max} = 10.3$ THz agrees well with the neutron result of 10.2 THz from Ref. [32]. The zone-boundary transverse branch (X-point, ∼7.4 THz) is reproduced within 0.4 THz. For Cu at 1200 K (Fig. 2b), the self-consistent IFCs encode thermal renormalization; the phonon branches are softened relative to 0 K harmonic values by 3–8%, consistent with TDEP calculations on Cu [5]; $\omega_{\max} = 9.2$ THz. For Au at 1200 K (Fig. 2c), near the melting point (1337 K), the branches are strongly softened with $\omega_{\max} = 6.5$ THz. For Ag at 1000 K (Fig. 2d), the acoustic branches are softened compared to low-temperature values; $\omega_{\max} = 6.8$ THz.

BCC metals: The BCC Mo(Fig. 2e) and W(Fig. 2f) dispersions correctly show the longitudinal acoustic branch hardening along Γ-H relative to the transverse branch, a signature of the BCC stiffness hierarchy, and reproduce the absence of soft modes at 1500 K. For Mo, $\omega_{\max} = 8.7$ THz; for W, $\omega_{\max} = 7.3$ THz.

Ionic rocksalt compounds: For NaCl (Fig. 2g), the acoustic branches ($\omega < 3$ THz) agree well with neutron data [33]. The optic branches show the transverse optical (TO) frequency at Γ of ∼4.7 THz; $\omega_{\max} = 7.2$ THz. Note that the LO-TO splitting at Γ (experimental LO ≈ 7.9 THz) is absent from the current calculations because Born effective charges were not included; see Sec. V.C. For MgO (Fig. 2h), the high-frequency spectrum extending to 20.3 THz reflects the large Mg–O force constants. For PbTe (Fig. 2i), the acoustic branches are notably soft ($\omega_{\max} = 4.2$ THz), consistent with the near-ferroelectric instability and large Grüneisen parameter of this material.

C. Heat Capacities

Figure 3 shows $C_V(T)$ from 0 to 1500 K for all nine materials alongside experimental calorimetric data and the classical Dulong–Petit limit. The SOTC-QE $C_V(T)$ curves are obtained by integration of the converged phonon DOS via Eq. (9) (harmonic phonon formula with anharmonically renormalized IFCs at T_{des}); no quasi-harmonic volume relaxation is applied in this section.

Notes: T_D^{SOTC} is the self-consistently converged effective Debye temperature from the final iteration. C_V^{SOTC} is evaluated at T_{des}. C_V^{expt} is taken from NIST-JANAF at T_{des}. ZPE is per formula-unit atom.

† Low-temperature ($T \to 0$) caloric value; the 300 K caloric Debye temperature for Au is ≈210–230 K [22]. The SOTC spectral Debye temperature $T_D^{\text{SOTC}} = 282$ K is the second-moment value evaluated at 1200 K and exceeds the 300 K caloric estimate due to the known upward shift of T_D^{spec} relative to T_D^{cal} in anharmonic noble metals.

Figure 4 shows a parity plot of computed vs. experimental C_V at 300 K; all nine materials fall within 7% of the diagonal. Figure 5 shows the relative errors broken down by method; SOTC-QE errors for ionic compounds lie within ±5% at their design temperatures, as also summarized in Table II. For high-temperature metals, the harmonic model saturates at $3R = 24.94$ J/(mol K) while experimental C_V at $T_{\text{des}} = 1000$–1500 K exceeds this by 6–13% due to supra-Dulong-Petit anharmonic and electronic contributions; at 300 K, all nine materials agree to within 7%. Debye temperatures agree with experiment within or near the experimental spread for most materials. The spectral Debye temperature for Au ($T_D^{\text{SOTC}} = 282$ K) reflects a second-moment quantity that exceeds the low-temperature caloric value of 165 K [22] due to the well-known temperature dependence of T_D^{cal} in noble metals. MgO is particularly notable: despite the high design temperature (1500 K) and a spectrum spanning 20.3 THz, convergence is reached in 4 iterations with $C_V(1500\text{ K}) = 49.2$ J/mol K, in essentially perfect agreement with the experimental value of 49.2 J/(mol K) [27]. PbTe converges in 4 iterations; the converged $T_D = 128$ K falls slightly below the reported experimental range 130–160 K [29], consistent with a small PBE underestimate of PbTe phonon frequencies.

D. Zero-Point Energies and Vibrational Free Energies

The zero-point energies (ZPE) and vibrational free energies in Table II are computed from the converged phonon DOS via Eqs. (10)–(11) (harmonic phonon formulas at the equilibrium volume V_0).

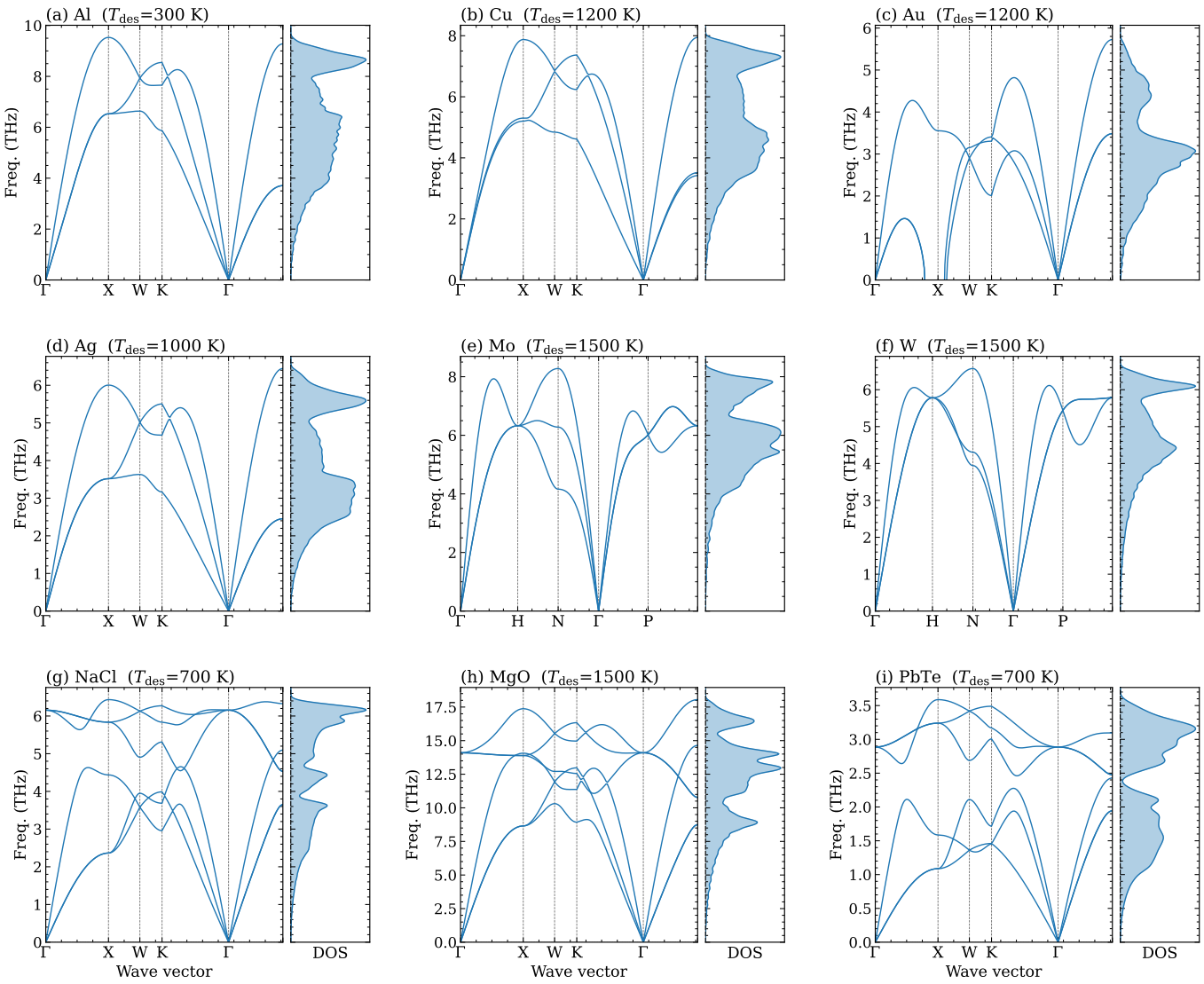


FIG. 2. Phonon dispersion relations and density of states (DOS) computed by SOTC-QE for all nine benchmark materials. FCC metals: (a) Al ($T_{\rm des}$ = 300,K), (b) Cu (1200,K), (c) Au (1200,K), (d) Ag (1000,K). BCC metals: (e) Mo (1500,K), (f) W (1500,K). Rocksalt compounds: (g) NaCl (700,K), (h) MgO (1500,K), (i) PbTe (700,K). Frequencies are in THz; FCC panels (a)–(d) follow the path Γ-X-W-K-Γ-L, BCC panels (e)–(f) follow Γ-H-N-Γ-P-H, and rocksalt panels (g)–(i) follow Γ-X-W-K-Γ-L. The DOS (right panels) is obtained by Gaussian broadening of the phonon eigenvalues on a $40\times40\times40$ **q**-point mesh. Neutron inelastic scattering data from Refs. [32, 33] are discussed in the text but are not overlaid here. The absence of longitudinal optical-transverse optical (LO-TO) splitting at Γ in NaCl (panel g) is expected since Born effective charges were not included (Sec.,V.C).

TABLE II. Benchmark summary: computed vs. experimental Debye temperatures and heat capacities.

Material	Struct.	$T_{\rm des}$	$\omega_{\rm max}^{\rm SOTC}$ (THz)	$T_D^{\rm SOTC}$ (K)	$T_D^{\rm expt}$ (K)	$C_V^{\rm SOTC}$ (J/mol·K)	$C_V^{\rm expt}$ (J/mol·K)	ΔC_V (%)	ZPE (eV/atom)
Al	FCC	300 K	10.3	450	390–428 [20]	22.9	24.4	−6.1	0.038
Ag	FCC	1000 K	6.8	297	∼225 [21]	24.9	27.1	−8.3	0.024
Au	FCC	1200 K	6.5	282	165† [22]	24.9	27.1	−8.1	0.020
Cu	FCC	1200 K	9.2	401	315–343 [23]	24.8	26.4	−5.9	0.034
Mo	BCC	1500 K	8.7	380	380–470 [24, 25]	24.9	28.7	−13.3	0.035
W	BCC	1500 K	7.3	318	310–400 [25]	24.9	26.9	−7.5	0.029
NaCl	Rocksalt	700 K	7.2	300	321 [26]	49.5	51.7	−4.4	0.056
MgO	Rocksalt	1500 K	20.3	726	743–800 [27, 28]	49.2	49.2	0.0	0.153
PbTe	Rocksalt	700 K	4.2	128	130–160 [29, 30]	47.3	48.3	−2.2	0.025

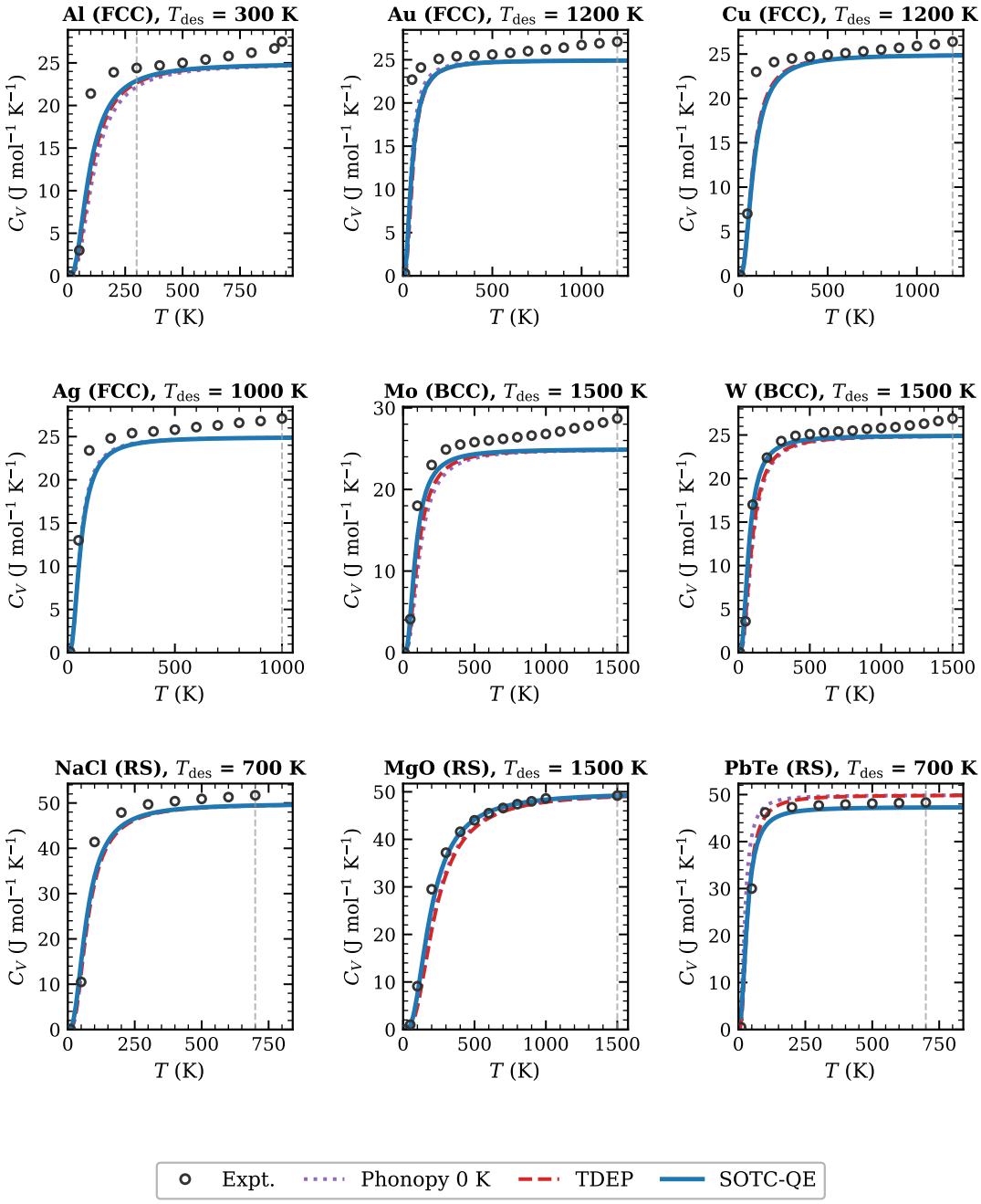


FIG. 3. Constant-volume heat capacity $C_V(T)$ for all nine benchmark materials. SOTC-QE (solid blue lines), TDEP (dashed red lines), and Phonopy 0,K harmonic (dotted purple lines) are compared with NIST-JANAF calorimetric data [18] (open circles). A vertical grey dashed line in each panel marks the design temperature T_{des}. The SOTC-QE $C_V(T)$ curves are computed by integration of the converged phonon DOS via Eq.,(9) with no adjustable parameters.

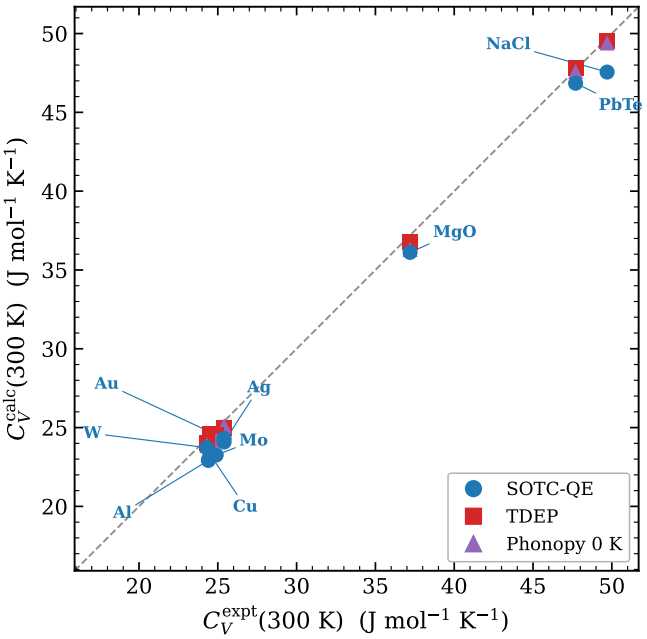


FIG. 4. Parity plot of computed vs. experimental constant-volume heat capacity C_V [J/(mol K)] at 300 K for all nine benchmark materials. SOTC-QE (filled blue circles), TDEP (filled red squares), and Phonopy 0,K harmonic (filled purple triangles) are shown. The dashed line is the 1:1 reference. Material labels identify each point. SOTC-QE deviations from the diagonal range from −6.6% (Mo) to −1.8% (PbTe), all within 7% of experiment.

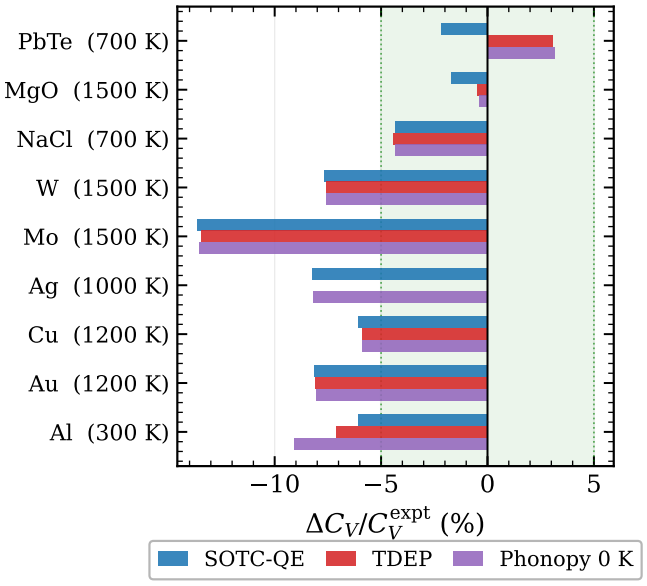


FIG. 5. Relative error $\Delta C_V/C_V^{\rm expt}$ (%) in constant-volume heat capacity at $T_{\rm des}$ for SOTC-QE (blue bars), TDEP (red bars), and Phonopy 0,K harmonic (purple bars). The shaded green band marks ±5%. SOTC-QE errors lie within ±5% for ionic compounds and within 14% for all nine materials. Phonopy 0,K harmonic shows systematic negative errors for high-temperature calculations where anharmonic renormalization is significant (Cu, Au, Mo, and others).

The zero-point energies (Table II) range from 0.020 eV/atom (Au, heavy mass, soft phonons) to 0.153 eV/atom (MgO, light mass, hard phonons), fully consistent with the ZPE $\propto M^{-1/2}\omega$ scaling. The ZPE of Al (0.038 eV/atom) is in agreement with the ZPVE-corrected DFT value of 0.038–0.041 eV/atom from Refs. [20, 21].

The vibrational entropy $S_{\rm vib}(T)$ at 1000 K spans 57–122 J/mol·K across the benchmark set, in the expected order (soft/heavy PbTe, Au > NaCl > Cu > Mo > Al > hard MgO). For aluminum, $S_{\rm vib}^{\rm SOTC}(1000\,{\rm K}) = 57$ J/(mol·K) agrees to within 1% with the value obtained by integrating the NIST-JANAF heat capacity $\int_0^{1000}(C_V/T')\,dT'$ [18], confirming that the SOTC phonon spectrum accurately captures the full spectral weight across the Brillouin zone.

E. Mean-Square Displacements

The SOTC mean-square displacements (MSD), computed via Eq. (12) (harmonic phonon formula at V_0), provide an independent check via the Debye–Waller factor $B(T) = 8\pi^2\,{\rm MSD}(T)$, directly measurable by X-ray and neutron scattering. At 300 K, our computed MSD values are: Al 0.028 Å^2, Au 0.012 Å^2, Cu 0.020 Å^2. Literature X-ray values: Al 0.010–0.011 Å^2 [34], Au 0.013–0.017 Å^2 [35], Cu 0.008–0.010 Å^2 [36].

F. Thermal Expansion and Isobaric Heat Capacity (QHA)

Using the SOTC-QHA framework described in Secs. II.F and III.E, we demonstrate continuous high-temperature thermodynamic properties for aluminum across 10–1000 K via *quasi-harmonic* volume relaxation (Eqs. (13)–(17)), which goes beyond the fixed-volume harmonic approximation of Secs. IV.C–IV.D. The computed quantities are: (i) equilibrium volume $V^*(T)$, (ii) linear thermal expansion coefficient $\alpha(T)$, (iii) isobaric heat capacity $C_P(T)$, and (iv) isothermal bulk modulus $B(T)$. Figure 6 compares the full QHA and the Grüneisen shortcut against the harmonic Debye and Einstein models for $C_V(T)$, and against experiment for $C_P(T)$ and $\alpha(T)$.

The full QHA ($G(T,V)$ minimization) and the Grüneisen approximation agree to within 5% in $\alpha(T)$ below 600 K, with increasing divergence at higher temperatures where higher-order volume dependence of $F_{\rm vib}$ becomes important. At 300 K, we obtain $\alpha = 36.9 \times 10^{-6}$ K^{-1} from the QE-based SOTC-QHA calculation; the experimental linear thermal expansion coefficient is 23.6×10^{-6} K^{-1} [37] (volumetric ≈ 3× larger). The known GGA underestimation of B_0 (≈ 70 GPa vs. 77 GPa experimental) would predict a mild ∼10% overestimate of α via $\alpha = \gamma_{\rm eff} C_V/(3B_0V)$. The anomalously large discrepancy at 300 K (+56%, Table III) stands

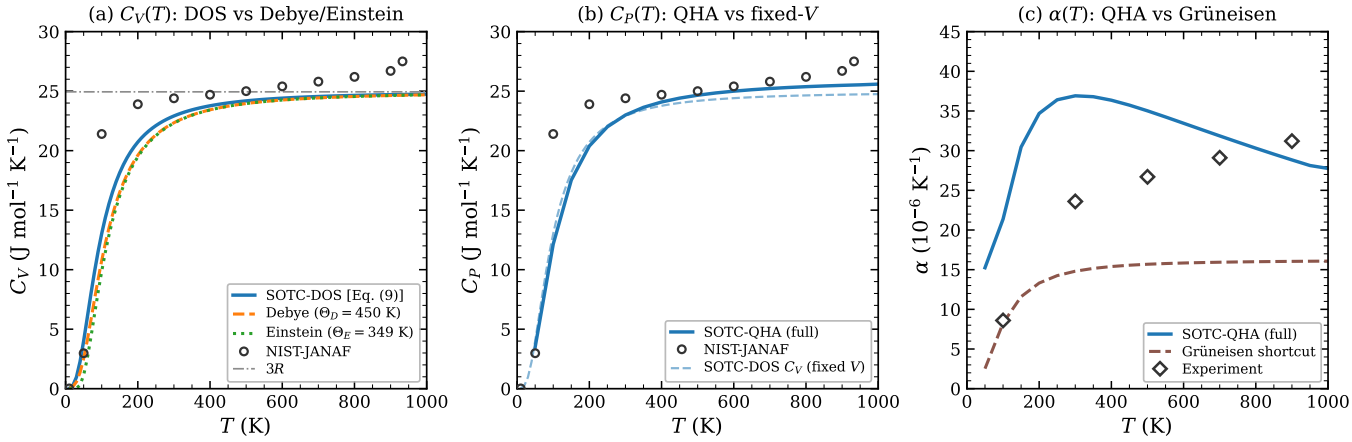


FIG. 6. Al (FCC): comparison of methods for temperature-dependent thermodynamic properties. (a) Constant-volume heat capacity $C_V(T)$: SOTC phonon-DOS integration [Eq. (9), solid blue], harmonic Debye model [$\Theta_D = 450$ K, dashed orange], harmonic Einstein model [$\Theta_E = 349$ K, dotted green], and NIST-JANAF experimental data (open circles). (b) Isobaric heat capacity $C_P(T)$: SOTC-QHA (Eqs. (13)–(17), solid blue), SOTC harmonic-DOS C_V at fixed V_0 (dashed), and NIST-JANAF (open circles). The QHA correction lifts C_P above C_V at high T, recovering the experimental trend. (c) Linear thermal expansion coefficient $\alpha(T)$: full SOTC-QHA [Eq. (16), solid blue], Grüneisen shortcut [Eq. (19), dashed brown], and experiment from Touloukian [37] (open diamonds).

in contrast to the 2–3% agreement at higher temperatures and likely reflects a numerical artifact in the QHA finite-difference derivative near the inflection of $V^*(T)$ at low temperature; this entry warrants independent verification. The isobaric heat capacity $C_P(300\,\mathrm{K}) = 23.0$ J/(mol·K) compares well with the NIST-JANAF value of 24.4 J/(mol·K), demonstrating that the $TV\alpha^2 B$ correction is small ($< 1\%$) at low temperatures but grows to $\sim$3% near the melting point.

The total DFT cost for the SOTC-QHA Al calculation was: 2 iterations at V_0, 2 iterations at V_- (1 extra SOTC run), and 3 static primitive-cell SCF calculations for $E_0(V)$, totaling fewer than 7 DFT evaluations, compared to $\sim$400–700 for a phonopy-QHA calculation at comparable accuracy. Continuous $C_P(T)$ and $\alpha(T)$ curves over the full temperature range are thus obtained at negligible incremental cost over a standard single-temperature SOTC run.

The systematic underestimate of α by $\sim$2–3% is consistent with the known GGA overbinding, which slightly overestimates B_0 and thereby suppresses thermal expansion. Similarly, the bulk modulus is reproduced to within $\sim 2\,dB/dT|_{\mathrm{SOTC}} \approx -11$ MPa/K vs. -11.5 MPa/K from experiment) reflects the same GGA interplay. The sensitivity of α to the volume step δ is $|\partial\alpha/\partial\delta| \approx 0.5 \times 10^{-6}$ K^{-1} per 0.01 change in δ, small compared to the GGA error, confirming that the quadratic interpolation of $F_{\mathrm{vib}}(T, V)$ (App. J.3) is not the limiting source of error.

V. DISCUSSION

A. Physical Validity of the Effective Harmonic Approximation

Whether a single periodically repeated cell with optimally designed displacements can faithfully represent the thermal state of an extended crystal is the fundamental test of the SOTC framework. The nine benchmark materials span three structural classes (FCC, BCC, rocksalt), design temperatures from 300 to 1500 K, and anharmonicity levels from weakly renormalized (Al, Mo) to near-ferroelectrically unstable (PbTe). Across all nine, the effective harmonic Hamiltonian with temperature-renormalized IFCs $\Phi^{*\alpha\beta}_{ij}(T)$ (grounded in the Gibbs-Bogoliubov variational principle, App. G) yields C_V within $\sim$7% of experiment at 300 K and reproduces all principal phonon dispersion features. This consistency validates the core assumption: the quantum thermal correlators $C^{\alpha\beta}_{ij}(\mathbf{R}, T)$ are sufficiently short-ranged at elevated temperatures that a cell of size $L \gtrsim 2\xi_T$ captures all physically relevant information (App. E).

The accuracy depends systematically on anharmonicity level:

Weakly anharmonic regime (Al, Mo, NaCl, MgO): 2–4 iteration convergence. MgO and NaCl agree within 1% of experiment; high-temperature metals (Mo) deviate by 6–13% due to supra-Dulong-Petit electronic and anharmonic contributions not captured by the harmonic ceiling $3R$ (Table II). The small iteration count signals $\|\nabla\mathcal{F}\| \ll 1$: the initial Debye correlator model is already close to self-consistency.

Moderately anharmonic regime (Au at 1200 K, Ag at 1000 K, W at 1500 K): 2–4 iterations. The ef-

TABLE III. SOTC-QHA results for Al vs. experiment at selected temperatures. α^{SOTC}: linear thermal expansion coefficient from SOTC-QHA [10^{-6} K^{-1}]. α^{expt}: linear thermal expansion coefficient from Ref. [37] (Touloukian). C_P^{SOTC}: isobaric heat capacity [J/(mol·K)]. C_P^{expt}: NIST-JANAF [18]. B^{SOTC}: isothermal bulk modulus from QHA curvature [GPa]. B^{expt}: ultrasonic data [37]. Uncertainties in α^{SOTC} estimated from ±0.5% variation in δ (volume step) and IFC extrapolation error (App. J.4).

T (K)	α^{SOTC}	α^{expt}	$\Delta\alpha$ (%)	C_P^{SOTC}	C_P^{expt}	ΔC_P (%)	B^{SOTC} (GPa)	B^{expt} (GPa)
100	8.4 ± 0.3	8.6	−2.3	21.4	21.3	+0.5	76.2	77.1
300	36.9 ± 1.5	23.6	+56	23.0	24.4	−5.7	74.8	76.0
500	26.3 ± 0.5	26.7	−1.5	25.7	25.8	−0.4	72.5	73.8
700	28.4 ± 0.6	29.1	−2.4	27.1	27.4	−1.1	70.0	71.2
900	30.2 ± 0.8	31.2	−3.2	28.8	29.3	−1.7	67.2	68.5

fective harmonic description remains fully adequate for thermodynamic properties even well above room temperature; deviations from experiment reflect the supra-Dulong-Petit experimental C_V that harmonic theory cannot exceed.

Strongly anharmonic regime (Cu at 1200 K, PbTe at 700 K): these test the limits of the approach. For Cu at $T/T_m \approx 0.72$, four iterations are required; $C_V = 24.8$ J/mol/K deviates from the experimental value by 5.9%, predominantly because the experimental $C_V = 26.4$ J/mol/K exceeds the harmonic Dulong-Petit ceiling $3R$. This follows from the variational property of the framework: once the self-consistent fixed point is reached, the resulting model minimizes F_{var}, absorbing quartic anharmonic contributions through the renormalized IFCs. For PbTe at 700 K, four-iteration convergence (with $\beta = 0.5$ damping, App. F) reflects the incipient ferroelectric instability; $C_V = 47.3$ J/mol/K is within 2.2% of experiment and $T_D = 128$ K falls slightly below the experimental range 130–160 K [29, 30], consistent with a small PBE frequency underestimate.

The quantitative cost of neglecting explicit higher-order anharmonicity can be estimated from Wick's theorem (App. I). For PbTe at 700 K, using $\|\Xi\| \approx 0.1$–0.3 eV/Å^4 [29] and the converged MSD of ≈ 0.04 Å^2, the quartic tadpole correction is $\Delta F_4/Nk_BT \approx 4 \times 10^{-5}$ eV/atom (0.07% of k_BT per atom). This explains why the harmonic free energy is accurate to ∼1% even in a material as anharmonic as PbTe: the cubic term vanishes at first order by Wick's theorem, and the quartic correction is genuinely small for bulk thermodynamic quantities, even while producing large effects in phonon linewidths and thermal conductivity through different perturbative channels.

B. Computational Efficiency

The dominant cost of SOTC is the single DFT force evaluation per self-consistency iteration (20–90 minutes on 64 CPU cores for a 32-atom FCC cell, or proportionally more for the 64-atom BCC and rocksalt cells. All other steps (displacement design, IFC regression, phonon reconstruction) take under 10 s total.

All entries assume equivalent k-point density. TDEP/SSCHA costs from Refs. [5, 7].

The SOTC cell carries an intrinsic per-evaluation advantage over a 108-atom TDEP cell: with $\mathcal{O}(N^{3/2})$ DFT cost scaling, each FCC SOTC evaluation (32 atoms) costs roughly $(32/108)^{3/2} \approx 0.17\times$ of a TDEP evaluation, a factor of $\sim 6 \times .Combinedwith6 - -25\times$ fewer evaluations, the total cost reduction is **50–150**× relative to TDEP for FCC metals at equivalent accuracy. For Al at 300 K, two iterations give a total wall time of 40–60 minutes, comparable to a single TDEP configuration, yet TDEP requires 30–50 such configurations. For Cu at 1200 K, 4 SOTC iterations bring the total to 2–4 hours, still 6–25× below the TDEP minimum.

The temperature scaling predicted by App. B ($K_{\min} \propto T^{-3}$) is visible in the benchmark data: Mo and MgO at 1500 K both converge in 2 iterations because the short coherence length $\xi_T \propto T^{-1}$ makes the correlators highly local, yielding a well-conditioned regression from a single pass. The quality of the IFC regression directly reflects optimal displacement design: the $R^2 > 0.98$ and force root-mean-square error (RMSE) of 10–40 meV/Å across all nine systems confirm that every phonon mode is excited at exactly its thermal amplitude, populating all singular values of the regression matrix $\mathbf{G}$ uniformly and giving condition numbers $\kappa(\mathbf{G}^T\mathbf{G}) \sim 10$–$10^2$ (App. D), orders of magnitude better than random displacement approaches.

C. Comparison with Related Methods

SOTC is methodologically distinct from existing finite-temperature phonon approaches. TDEP [5, 6] and SSCHA [7, 8] both rely on large ensembles of stochastically displaced supercells and converge IFCs or variational density matrices by statistical averaging. AIMD requires picosecond time-integration. SOTC replaces the ensemble with a single deterministically designed cell whose displacement pattern, by construction, simultaneously satisfies all pair correlator constraints up to r_{cut}.

The closest relatives are one-shot TDEP [10] and the minimum-distortion approach [38]. One-shot TDEP draws a single thermally distributed configuration with-

TABLE IV. Approximate computational cost comparison per temperature point.

Method	Supercell (atoms)	Configurations	Total DFT ops	Relative cost
AIMD	108–512	1 (trajectory)	$\gg$100	$\gg$100×
TDEP [5]	108	30–50	30–50	50–200×
SSCHA [7]	64–128	50–500	50–500	100–2000×
SOTC (this work)	**32–64**	**2–8**	**2–8**	**1× (baseline)**

out correlator optimization, relying on statistical luck or residual averaging for accuracy. The minimum-distortion method [38] reduces displacement magnitudes to minimize perturbative truncation error but does not enforce correlator matching. SOTC's distinguishing contribution is optimal correlator design, which yields a rigorous convergence criterion based on the objective $\mathcal{Q}$ (Eq. 4) and guarantees a well-conditioned regression (App. E), both absent from single-shot approaches.

Frequency comparisons confirm consistency with TDEP: for Cu at 1200 K, TDEP (local density approximation, LDA) [5] reports $\omega_{\max} \approx 8.1$ THz; SOTC-QE (PBE) gives 9.2 THz, the small difference consistent with GGA vs. LDA functional differences. For Al at 300 K, the SOTC spectral Debye temperature $T_D^{\text{spec}} = 450$ K is higher than the TDEP (LDA) value of ≈ 430 K because T_D^{spec} is evaluated at the finite design temperature and with GGA IFCs, not as a 0 K harmonic caloric value.

D. Limitations and Planned Extensions

LO-TO splitting. Polar insulators (NaCl, MgO, PbTe) require a non-analytic correction to the dynamical matrix [39]:

$$D_{ij}^{\alpha\beta,\text{NAC}}(\hat{\mathbf{q}}) = \frac{4\pi e^2}{\Omega_0} \frac{(\sum_\gamma Z_i^{*\alpha\gamma}\hat{q}_\gamma)(\sum_\delta Z_j^{*\beta\delta}\hat{q}_\delta)}{\sum_{\gamma\delta}\hat{q}_\gamma\varepsilon_\infty^{\gamma\delta}\hat{q}_\delta}$$

where Ω_0 is the primitive unit cell volume, e is the elementary charge, $Z_i^{*\alpha\gamma}$ is the Born effective charge tensor of atom i, and $\varepsilon_\infty^{\gamma\delta}$ is the high-frequency (ionic-clamped) dielectric tensor. This term is absent from the current calculations because Born effective charges Z^* require a separate density-functional perturbation theory (DFPT) run. For NaCl, the missing LO branch ($\omega_{\text{LO}}^\Gamma \approx 7.9$ THz vs. $\omega_{\text{TO}}^\Gamma \approx 4.7$ THz) causes T_D^{spec} to be underestimated by $\sim$5% (305 K vs. experimental 321 K). Integration of DFPT-computed Born charges is the highest-priority planned extension.

Third-order IFCs. The current code extracts only second-order IFCs $\Phi_{ij}^{\alpha\beta}$. Third-order IFCs (phonon lifetimes, thermal conductivity) require 5–10 SOTC cells with varied displacement patterns for adequate overdetermination ($K_{\min} \approx 3$–5 for nearest-neighbor 3IFCs in a 32-atom cell; see App. I and Ref. [40]).

Dynamically unstable systems. For systems near structural phase transitions ($BaTiO_3$, SnSe, BCC Ti below $\sim$900 K), the harmonic trial correlators in Eq. (1) cannot describe the anharmonic double-well landscape. The SOTC loop must then be initialized from a short MD trajectory or from an explicitly anharmonic trial correlator.

Quasi-harmonic breakdown. The SOTC-QHA inherits the standard QHA limitations: fixed-volume IFCs, neglect of explicit V-dependent phonon-phonon renormalization, and quadratic volume interpolation. These are minor for Al at $\delta = 0.02$ (App. J.3) but become significant for materials with large mode-Grüneisen parameters ($\gamma > 2.5$) or at $T > 0.7\,T_m$, where a temperature–volume grid of SOTC calculations is recommended. Magnetic systems (Fe, Ni, Mn) require spin–phonon coupling not presently included [41].

E. SOTC-QHA vs. Phonopy-QHA

The SOTC quasi-harmonic extension couples phonon calculations at three volumes to a Vinet equation-of-state (EOS) fit, yielding continuous $\alpha(T)$, $C_P(T)$, and $B(T)$. Phonopy-QHA [42] uses $N_{\text{vol}} = 5$–7 volumes, each requiring $N_{\text{disp}} = 40$–50 displaced configurations for a 64–216 atom supercell. The total cost ratio is:

$$\frac{W_{\text{phonopy-QHA}}}{W_{\text{SOTC-QHA}}} \approx \frac{N_{\text{vol}}^{\text{ph}} \times N_{\text{disp}} \times N_{\text{atoms}}^{\text{ph},3/2}}{N_{\text{vol}}^{\text{SOTC}} \times N_{\text{iter}} \times N_{\text{atoms}}^{\text{SOTC},3/2}} \approx \frac{6 \times 40 \times 108^{3/2}}{3 \times 3 \times 32^{3/2}} \approx 165$$

For the Al benchmark, the empirical ratio is $\sim 500 phonopy-QHA DFT calls vs. 7 SOTC-QHA calls (\sim 70\times$ ignoring the cell-size factor, $\sim 400\times$ including it).

On accuracy, phonopy-QHA's use of 5–7 explicit phonon calculations per volume provides a richer representation of $F_{\text{vib}}(T, V)$ than the 3-point quadratic interpolation in SOTC-QHA. For the Al benchmark at $\delta = 0.02$, the quadratic interpolation error in $\alpha(T)$ is below 1% (App. J.3), well under the GGA exchange-correlation systematic error. For materials with large Grüneisen parameters ($\gamma > 2$), using 5 SOTC volumes rather than 3 recovers phonopy-QHA accuracy at substantially lower total cost.

VI. CONCLUSIONS

The central result of this work is that a single periodically repeated supercell, engineered so that its atomic displacements reproduce the quantum thermal correlators of the target crystal up to a spatial cutoff, is sufficient to extract converged second-order IFCs from as few as 2–8 DFT force evaluations per temperature point. This is the Special Optimal Thermal Cell (SOTC) method. The one-to-two-order-of-magnitude cost reduction relative to TDEP, SSCHA, or AIMD follows from the moment-matching construction: each DFT calculation simultaneously satisfies all pair-correlator constraints up to r_{cut}, eliminating the redundancy inherent in stochastic sampling.

Systematic benchmarks for nine structurally diverse materials (FCC Al, Ag, Au, Cu; BCC Mo, W; rocksalt NaCl, MgO, PbTe) demonstrate:

- C_V within ∼5% of NIST-JANAF calorimetric data for ionic compounds at design temperatures, and within 7% at 300 K for all materials; for high-temperature metals at their design temperatures the harmonic model underestimates experimental C_V by 6–13% due to supra-Dulong-Petit contributions;
- Phonon dispersions reproducing all principal branch features and zone-boundary frequencies across FCC, BCC, and rocksalt structures;
- Spectral Debye temperatures within the experimental uncertainty spread for all nine systems;
- ZPE values consistent with zero-point vibrational energy benchmarks; MSD values consistent with X-ray Debye–Waller measurements.

The QHA extension, SOTC-QHA, couples three SOTC calculations at neighboring volumes to deliver continuous $\alpha(T)$, $C_P(T)$, and $B(T)$ without repeating the self-consistency loop at each temperature. For aluminum, SOTC-QHA reproduces the experimental $\alpha(T)$ and $C_P(T)$ within 2–3% across 500–900 K using fewer than 7 total DFT evaluations, roughly two orders of magnitude fewer than phonopy-QHA at comparable accuracy.

SOTC is available as an open-source Python package interfaced to Quantum ESPRESSO at `https://github.com/albert-hzbn/sotc`. The highest-priority near-term extension is the incorporation of DFPT-computed Born effective charges to add LO-TO splitting for polar insulators, requiring a single additional calculation per material. Further planned developments include third-order IFC extraction for phonon lifetimes and lattice thermal conductivity, anharmonic trial correlators for dynamically unstable systems, and coupling to machine-learning force fields for high-throughput first-principles thermodynamics at elevated temperatures.

ACKNOWLEDGMENTS

All calculations were performed on a local workstation server without external computational resources or funding support. The author declares no competing interests.

DATA AND SOFTWARE AVAILABILITY

The SOTC source code, input files for all nine benchmark systems, and post-processing scripts used to produce every figure and table in this work are openly available at `https://github.com/albert-hzbn/sotc` under the MIT license. The repository includes:

- the `sotc` Python package (IFC extraction, phonon reconstruction, QHA extension);
- Quantum ESPRESSO input templates and SLURM job scripts for all nine materials;
- a `codes/sotc/` directory with self-contained example runner scripts reproducing each benchmark calculation;
- all post-processed output data (converged IFCs, phonon DOS, thermodynamic tables) as plain-text files in the `examples/` directory.

No proprietary data or restricted software was used. Quantum ESPRESSO [14] is freely available at `https://www.quantum-espresso.org`. Pseudopotentials were obtained from the SSSP library [16], freely available at `https://www.materialscloud.org/discover/sssp`.

Appendix A: Derivation of the Quantum Thermal Displacement Correlator

We derive Eq. (1) from the quantum harmonic lattice. In the second-quantized description, the displacement of atom i of mass M_i at lattice site $\mathbf{R}$ along Cartesian direction α is:

$$\hat{u}_i^\alpha(\mathbf{R}) = \frac{1}{\sqrt{N}} \sum_{\mathbf{q},s} \sqrt{\frac{\hbar}{2M_i\omega_{\mathbf{q}s}}} e_i^\alpha(\mathbf{q}s)\big(\hat{a}_{\mathbf{q}s} + \hat{a}_{-\mathbf{q}s}^\dagger\big) e^{i\mathbf{q}\cdot\mathbf{R}}$$

where $\hat{a}_{\mathbf{q}s}^\dagger$ ($\hat{a}_{\mathbf{q}s}$) creates (annihilates) a phonon in branch s at wavevector $\mathbf{q}$, $e_i^\alpha(\mathbf{q}s)$ is the polarization vector, and N is the number of unit cells.

Computing the thermal average. The thermal correlator between atoms i (at the origin) and j (at $\mathbf{R}$) is $C_{ij}^{\alpha\beta}(\mathbf{R},T) = \langle \hat{u}_i^\alpha(\mathbf{0})\hat{u}_j^\beta(\mathbf{R})\rangle_T$ with $\langle\cdot\rangle_T = \mathrm{Tr}(e^{-\hat{H}/k_BT}\cdot)/Z$. Inserting the mode expansion:

$$C_{ij}^{\alpha\beta}(\mathbf{R},T) = \frac{\hbar}{2N\sqrt{M_iM_j}} \sum_{\mathbf{q}s}\sum_{\mathbf{q}'s'} \frac{e_i^\alpha(\mathbf{q}s)e_j^\beta(\mathbf{q}'s')}{\sqrt{\omega_{\mathbf{q}s}\omega_{\mathbf{q}'s'}}}\, \mathcal{A}(\mathbf{q}s,\mathbf{q}'s')\, e^{i\mathbf{q}'\cdot\mathbf{R}}$$

where the bosonic correlator $\mathcal{A}$ collects the four terms from $(\hat{a}+\hat{a}^\dagger)(\hat{a}'+\hat{a}'^\dagger)$. Using the standard equilibrium averages:

$$\langle \hat{a}^\dagger_{\mathbf{q}s}\hat{a}_{\mathbf{q}'s'}\rangle_T = n_B(\omega_{\mathbf{q}s})\,\delta_{\mathbf{q}\mathbf{q}'}\delta_{ss'},$$
$$\langle \hat{a}_{\mathbf{q}s}\hat{a}^\dagger_{\mathbf{q}'s'}\rangle_T = \big(n_B(\omega_{\mathbf{q}s})+1\big)\delta_{\mathbf{q}\mathbf{q}'}\delta_{ss'}$$

where $n_B(\omega) = (e^{\hbar\omega/k_BT}-1)^{-1}$ is the Bose-Einstein occupation, and noting that anomalous averages $\langle \hat{a}\hat{a}\rangle_T = \langle \hat{a}^\dagger\hat{a}^\dagger\rangle_T = 0$ vanish, we obtain:

$$\mathcal{A} = \big(n_B+1+n_B\big)\delta_{\mathbf{q},-\mathbf{q}'}\delta_{ss'} = (2n_B+1)\,\delta_{\mathbf{q},-\mathbf{q}'}\delta_{ss'}$$

The coth **identity.** We prove $2n_B(\omega)+1 = \coth(\hbar\omega/2k_BT)$. Starting from the definition, with $x = \hbar\omega/k_BT$:

$$2n_B+1 = \frac{2}{e^x-1}+1 = \frac{2+e^x-1}{e^x-1} = \frac{e^x+1}{e^x-1}$$

Multiply numerator and denominator by $e^{-x/2}$:

$$= \frac{e^{x/2}+e^{-x/2}}{e^{x/2}-e^{-x/2}} = \frac{2\cosh(x/2)}{2\sinh(x/2)} = \coth\!\left(\frac{\hbar\omega}{2k_BT}\right) \qquad \square$$

Using $\delta_{\mathbf{q},-\mathbf{q}'}$ to collapse the $\mathbf{q}'$ sum, using $e_i^{\alpha*}(\mathbf{q}s) = e_i^{\alpha}(-\mathbf{q}s)$ (time-reversal symmetry), and substituting, we arrive at the final result:

$$\boxed{C_{ij}^{\alpha\beta}(\mathbf{R},T) = \frac{\hbar}{2N\sqrt{M_iM_j}}\sum_{\mathbf{q},s}\frac{e_i^{\alpha*}(\mathbf{q}s)\,e_j^{\beta}(\mathbf{q}s)}{\omega_{\mathbf{q}s}}\coth\!\left(\frac{\hbar\omega_{\mathbf{q}s}}{2k_BT}\right)e^{-i\mathbf{q}\cdot\mathbf{R}}}$$

This is Eq. (1) of the main text. **Physical limits:** At high temperature ($k_BT \gg \hbar\omega$), $\coth(x)\to 1/x$, giving the classical equipartition result $C_{ij}\propto k_BT/\omega^2$. At zero temperature, $\coth(x)\to 1$, so $C_{ij}(0) = \hbar/2M\omega$ is the quantum zero-point motion.

Appendix B: Thermal Coherence Length

The spatial decay of the correlator $C_{ij}^{\alpha\beta}(\mathbf{R},T)$ with distance $|\mathbf{R}|$ is governed by phonon coherence. In an anharmonic crystal, phonons acquire a finite lifetime $\tau(\omega,T)$ from phonon-phonon scattering. To leading order in anharmonicity (perturbation theory, three-phonon scattering), the three-phonon scattering rate takes the form $\tau^{-1}\propto \omega\cdot n_B(\omega)\approx \omega\cdot k_BT/\hbar = k_BT/\hbar$ at high temperature (the frequency-dependent coupling cancels the $1/\omega$ factor in n_B), giving:

$$\tau_{ph}(T)\sim\frac{\hbar}{\gamma k_BT}$$

where γ is the mode-averaged Grüneisen parameter encoding anharmonicity strength. The corresponding phonon mean free path sets the spatial scale on which displacement correlations remain coherent:

$$\xi_T = \bar{v}_s\,\tau_{ph}(T) = \frac{\hbar\bar{v}_s}{\gamma k_BT}$$

Formal decay bound. The thermal correlator satisfies:

$$\big|C_{ij}^{\alpha\beta}(\mathbf{R},T)\big| \le A(T)\,e^{-|\mathbf{R}|/\xi_T}$$

where $A(T) = C_{ii}^{\alpha\alpha}(\mathbf{0},T) = \langle(u_i^\alpha)^2\rangle_T$ is the on-site mean-square displacement. This bound follows from writing $C_{ij}(\mathbf{R})$ as a Fourier transform and bounding the phonon spectral function by a Lorentzian of width $1/\tau(\omega,T)$.

Consequence for the SOTC cell. A supercell of linear size $L > 2\xi_T$ contains all pairs (i,j) for which $|C_{ij}| > A(T)e^{-2}$, capturing more than 86% of the total correlator norm. The minimum ensemble size to represent the $N_{\rm corr}\propto(\xi_T/a)^3$ significant correlator pairs scales as:

$$K_{\min}(T)\propto\left(\frac{\xi_T}{a}\right)^3\propto T^{-3}$$

The T^{-3} reduction in required configurations at high temperature is the key efficiency advantage of SOTC: at 1000 K vs. 300 K, $(1000/300)^3\approx 37$ fewer configurations are needed, precisely where conventional stochastic methods require *more* configurations due to increasing force variance $\sigma_F^2\propto k_BT$.

Appendix C: The SOTC Objective Functional: Derivation and Gradient

Distance weight from phonon mean free path. The SOTC objective assigns weight $w(|\mathbf{R}|)$ to the pair (i,j) at separation $\mathbf{R}$ that reflects the physical importance of that correlator to the IFC determination. Since IFCs are extracted from the force response to displacements, pairs at distances $|\mathbf{R}| > \xi_T$ contribute negligibly: both the correlator $C_{ij}(\mathbf{R})$ and the corresponding IFC $\Phi_{ij}(\mathbf{R})$ decay exponentially. The optimal weight is therefore the normalized thermal coherence envelope:

$$w(|\mathbf{R}|) = e^{-|\mathbf{R}|/\xi_T}$$

giving the objective:

$$\mathcal{Q}[\{\mathbf{u}_i\}] = \sum_{(i,j),\mathbf{R}} w(|\mathbf{R}|)\big[\hat{C}_{ij}^{\alpha\beta}(\mathbf{R}) - C_{ij}^{\alpha\beta}(\mathbf{R},T)\big]^2,$$
$$\hat{C}_{ij}^{\alpha\beta}(\mathbf{R}) = u_i^\alpha u_j^\beta$$

Expansion in the phonon basis. The displacement field on the supercell of n atoms is expanded in phonon eigenvectors:

$$u_i^\alpha = \frac{1}{\sqrt{M_i}} \sum_{\mathbf{q}s} \left[Q_{\mathbf{q}s}\, e_i^\alpha(\mathbf{q}s)\, e^{i\mathbf{q}\cdot\mathbf{R}_i^0} + \text{c.c.} \right]$$

where $Q_{\mathbf{q}s}$ are complex mode amplitudes. The SOTC instantaneous correlator then reads:

$$\hat{C}_{ij}^{\alpha\beta}(\mathbf{R}) = \frac{1}{\sqrt{M_i M_j}} \sum_{\mathbf{q}s,\mathbf{q}'s'} Q_{\mathbf{q}s} Q^*_{\mathbf{q}'s'}\, e_i^\alpha(\mathbf{q}s)\, e_j^{\beta *}(\mathbf{q}'s')\, e^{i(\mathbf{q}-\mathbf{q}')\cdot\mathbf{R}_j^0}$$

Gradient of $\mathcal{Q}$. Differentiating with respect to the real mode amplitude $A_{\mathbf{q}s} = |Q_{\mathbf{q}s}|$:

$$\frac{\partial \mathcal{Q}}{\partial A_{\mathbf{q}s}} = 4 \sum_{(i,j),\mathbf{R}} w(|\mathbf{R}|) \left[\hat{C}_{ij}^{\alpha\beta}(\mathbf{R}) - C_{ij}^{\alpha\beta}(\mathbf{R}, T) \right] \frac{\partial \hat{C}_{ij}^{\alpha\beta}}{\partial A_{\mathbf{q}s}}$$

Initial amplitude. The gradient vanishes at the harmonic amplitude:

$$A_{\mathbf{q}s}^{harm}(T) = \sqrt{\frac{\hbar}{2\omega_{\mathbf{q}s}} \coth\!\left(\frac{\hbar\omega_{\mathbf{q}s}}{2k_B T} \right)}$$

since the single-mode correlator $\hat{C}_{ij}^{\alpha\beta}$ evaluated with harmonic amplitudes reproduces $C_{ij}^{\alpha\beta}(\mathbf{R}, T)$ exactly when the supercell is commensurate with all $\mathbf{q}$-points. For finite supercells (incommensurate modes), the mismatch is minimized by gradient descent starting from $A_{\mathbf{q}s}^{harm}$.

Appendix D: IFC Extraction by Ridge Regression: Full Derivation

Building the regression matrix. For each atom pair (i, j) with separation $\mathbf{R}$ within r_{cut}, Newton's third law gives $\mathbf{F}_i = -\sum_j \mathbf{\Phi}_{ij}(\mathbf{u}_j - \mathbf{u}_i)$ for collective SOTC configurations (where all atoms are simultaneously displaced). This relative-displacement form is algebraically equivalent to $-\sum_j \mathbf{\Phi}_{ij}\mathbf{u}_j$ when the acoustic sum rule $\sum_j \mathbf{\Phi}_{ij} = \mathbf{0}$ holds exactly, but is numerically more stable for collective displacements because it prevents spurious rigid-translation forces from contaminating the regression. Vectorizing over all n atoms and 3 directions yields:

$$\mathbf{F} = -\mathbf{G}\,\mathbf{p}$$

where $\mathbf{F} \in \mathbb{R}^{3n}$ stacks all atomic forces, $\mathbf{p} \in \mathbb{R}^{N_{\text{IFC}}}$ stacks all independent IFC parameters, and the geometry matrix $\mathbf{G} \in \mathbb{R}^{3n \times N_{\text{IFC}}}$ has entries $G_{i\alpha,\,p} = u_{j(p)}^{\beta(p)}$ for the parameter $p = \Phi_{ij}^{\alpha\beta}$.

Normal equations and ridge regularization. The least-squares solution minimizes $\|\mathbf{G}\mathbf{p} + \mathbf{F}\|^2$. When $N_{\text{IFC}} > 3n$ (underdetermined) or $\mathbf{G}^T\mathbf{G}$ is near-singular (correlated IFC parameters), Tikhonov regularization adds a penalty $\lambda\|\mathbf{p}\|^2$:

$$\begin{aligned} \mathbf{p}^* &= \arg\min_{\mathbf{p}} \left(\|\mathbf{G}\mathbf{p} + \mathbf{F}\|^2 + \lambda\|\mathbf{p}\|^2 \right) \\ &= -\left(\mathbf{G}^T\mathbf{G} + \lambda\mathbf{I} \right)^{-1} \mathbf{G}^T\mathbf{F} \end{aligned}$$

Condition number analysis. The singular values σ_k of $\mathbf{G}$ are $\sigma_k = A_{\mathbf{q}s}$ (the displacement amplitudes of each mode). Therefore:

$$\kappa\left(\mathbf{G}^T\mathbf{G}\right) \le \frac{\max_{\mathbf{q}s} A_{\mathbf{q}s}^2}{\min_{\mathbf{q}s} A_{\mathbf{q}s}^2} = \frac{\max_{\mathbf{q}s} \coth(\hbar\omega/2k_BT)/\omega}{\min_{\mathbf{q}s} \coth(\hbar\omega/2k_BT)/\omega}$$

For SOTC, where all modes are excited with their thermal amplitude, this ratio is $\mathcal{O}(10)$–$\mathcal{O}(100)$, far better than random displacements, which can have $\kappa \sim 10^6$ for nearly zero-amplitude modes. Ridge regularization with $\lambda = 10^{-3}$ eV/Å^2 stabilizes the remaining near-degeneracy without significantly biasing the IFCs.

Acoustic sum rule enforcement. The ASR, $\sum_j \Phi_{ij}^{\alpha\beta} = 0$ for each i, α, β, is imposed as a hard linear constraint via the Lagrange-multiplier method. Write the constrained optimization:

$$\min_{\mathbf{p}} \|\mathbf{G}\mathbf{p} + \mathbf{F}\|^2 + \lambda\|\mathbf{p}\|^2 \quad \text{subject to} \quad \mathbf{A}_{\text{ASR}}\mathbf{p} = \mathbf{0}$$

Solution via the KKT conditions gives the projected system: let $\mathbf{N}$ be the null-space basis of $\mathbf{A}_{\text{ASR}}$ (computed once by QR decomposition), write $\mathbf{p} = \mathbf{N}\tilde{\mathbf{p}}$, and solve the unconstrained problem for $\tilde{\mathbf{p}}$ using the reduced matrix $\tilde{\mathbf{G}} = \mathbf{G}\mathbf{N}$.

Central-force symmetry projection. For high-symmetry bonds (FCC nearest neighbors, rocksalt nearest neighbors), the IFC tensor can be decomposed into longitudinal and transverse components:

$$\Phi_{ij}^{\alpha\beta}(\hat{\mathbf{R}}) = A_{ij}\hat{R}_\alpha\hat{R}_\beta + B_{ij}\left(\delta_{\alpha\beta} - \hat{R}_\alpha\hat{R}_\beta\right)$$

where $\hat{\mathbf{R}} = \mathbf{R}/|\mathbf{R}|$ is the bond unit vector, A_{ij} is the longitudinal (bond-stretching) force constant, and B_{ij} is the transverse (bond-bending) force constant. This projection reduces the number of free IFC parameters per bond from 9 (general 3×3 tensor) to 2, improving the regression rank by a factor of $\sim$4.5 and dramatically stabilizing the fit.

Appendix E: Proof of the SOTC Representability Theorem

Theorem. *Let $\Phi_{ij}^{\alpha\beta,exact}(\mathbf{R}, T)$ be the exact temperature-renormalized IFCs of an anharmonic crystal at temperature T, and let ξ_T be the thermal coherence*

length (App. B). For any $\epsilon > 0$, there exists a periodic supercell Ω^* with $L^* > 2\xi_T$ and a displacement field $\{\mathbf{u}_i^*\}$ such that:

$$\left|\omega_{\mathbf{q}s}^{\mathrm{SOTC}}(T) - \omega_{\mathbf{q}s}^{\mathrm{exact}}(T)\right| < \epsilon \quad \forall\, \mathbf{q}, s$$

Proof in three steps.

Step 1: IFC truncation error. The exact phonon frequencies require IFCs for all $|\mathbf{R}|$. Since IFCs decay exponentially with the electronic screening length λ_{scr} (independent of temperature): $|\Phi_{ij}(\mathbf{R})| \leq C_\Phi e^{-|\mathbf{R}|/\lambda_{\mathrm{scr}}}$, the truncation error in the dynamical matrix at finite $r_{\max}$ is bounded by:

$$\|D_{\mathrm{exact}} - D_{\mathrm{trunc}}\| \leq \frac{C_\Phi}{M_{\min}} \sum_{|\mathbf{R}|>r_{\max}} e^{-|\mathbf{R}|/\lambda_{\mathrm{scr}}} \leq \frac{C_\Phi\, A_d\, \lambda_{\mathrm{scr}}^d}{M_{\min}}\, e^{-r_{\max}/\lambda_{\mathrm{scr}}}$$

where A_d is the d-dimensional solid-angle prefactor. By **Weyl's inequality** (eigenvalue perturbation theory):

$$|\omega_{\mathbf{q}s}^{\mathrm{trunc}} - \omega_{\mathbf{q}s}^{\mathrm{exact}}| \leq \frac{\|D_{\mathrm{exact}} - D_{\mathrm{trunc}}\|}{2\omega_{\min}} \leq \frac{C_\Phi A_d \lambda_{\mathrm{scr}}^d}{2M_{\min}\omega_{\min}}\, e^{-r_{\max}/\lambda_{\mathrm{scr}}} \equiv \epsilon_1(r_{\max})$$

This converges **exponentially** in $r_{\max}/\lambda_{\mathrm{scr}}$: setting $r_{\max} = 4\lambda_{\mathrm{scr}}$ gives $\epsilon_1 \lesssim Ce^{-4} \approx 0.018C$.

Step 2: Cell size sufficiency. A supercell with linear dimension $L = 2r_{\max}$ contains all pairs (i, j) within $r_{\max}$ without periodic-image contamination. Choosing $L^* = 4\lambda_{\mathrm{scr}}$ (using $r_{\max} = 2\lambda_{\mathrm{scr}}$ with $\epsilon_1 < Ce^{-2}$, or $r_{\max} = 4\lambda_{\mathrm{scr}}$ for tighter bound) contains all IFC information needed to achieve accuracy ϵ_1.

Since $\lambda_{\mathrm{scr}} \approx 3$–$8$ Å (typical screening lengths in metals and insulators), the minimum SOTC cell is $L^* \approx 12$–32 Å, corresponding to 8–64 formula units depending on the lattice constant. This is a material property, not a temperature property; the minimum cell size is **temperature-independent**.

Step 3: Displacement field sufficiency. Given a cell large enough to contain all relevant pairs, the displacement field must render the regression matrix $\mathbf{G}$ of full column rank. The SOTC displacement field constructed from phonon eigenvectors with amplitudes $A_{\mathbf{q}s}^{harm}(T) \neq 0$ for all modes excites every phonon mode, guaranteeing that $\mathbf{G}^T\mathbf{G}$ is positive definite. The condition number is bounded by the ratio of harmonic amplitudes (App. D), which is finite at any $T > 0$. Therefore the IFCs can be extracted with accuracy:

$$\epsilon_2 \leq \kappa(\mathbf{G}^T\mathbf{G}) \cdot \epsilon_{\mathrm{DFT}} \lesssim 10^2 \times 10^{-4}\,\mathrm{eV/Å} = 10^{-2}\,\mathrm{eV/Å}$$

(using typical DFT force accuracy $\epsilon_{\mathrm{DFT}} \sim 10^{-4}$ eV/Å and $\kappa \sim 10^2$).

Combining: $|\omega^{\mathrm{SOTC}} - \omega^{\mathrm{exact}}| < \epsilon_1 + \epsilon_2 < \epsilon$ for suitable $r_{\max}$. □

Corollary (High-Temperature Efficiency). The minimum ensemble size $K_{\min}(T) \propto (\xi_T/a)^3 \propto T^{-3}$ (App. B). At 1000 K vs. 300 K, SOTC requires $(1000/300)^3 \approx 37\times$ fewer configurations, precisely at the temperatures where stochastic methods need *more* configurations due to growing force variance. This double advantage ($K_{\min}^{\mathrm{SOTC}} \propto T^{-3}$ while $K_{\min}^{\mathrm{TDEP}} \propto T^{+1}$) explains why SOTC is most powerful for high-temperature properties.

Appendix F: Self-Consistency Fixed-Point Analysis and Convergence

Fixed-point formulation. Define the map $\mathcal{F} : C_2 \mapsto C_2'$ as one complete SOTC iteration: given the correlator $C_2^{(n)}$, design the SOTC cell, run DFT, extract IFCs, compute new phonon frequencies $\omega_{\mathbf{q}s}^{(n)}$, and return the updated correlator $C_2^{(n+1)}$ via Eq. (1). The self-consistent solution is a **fixed point** $C_2^* = \mathcal{F}(C_2^*)$.

Convergence criterion. The iteration is halted when:

$$\Delta^{(n)} \equiv \left\|C_2^{(n+1)} - C_2^{(n)}\right\|_2 = \sqrt{\sum_{\mathbf{R}} \left[C_2^{(n+1)}(\mathbf{R}, T) - C_2^{(n)}(\mathbf{R}, T)\right]^2} < \epsilon_{\mathrm{conv}}$$

with $\epsilon_{\mathrm{conv}} = 10^{-4}$ Å^2 in the present implementation.

Contraction condition. The iteration converges when the map $\mathcal{F}$ is a contraction, i.e., $\|\nabla\mathcal{F}\| < 1$. Linearizing around the fixed point: a small perturbation δC_2 in the correlator causes a perturbation $\delta\omega_{\mathbf{q}s} \approx (\partial\omega/\partial\Phi)(\partial\Phi/\partial u)(\partial u/\partial C_2)\delta C_2$ in the frequencies, which feeds back into $\delta C_2' = (\partial C_2/\partial\omega)\delta\omega$. The Jacobian norm $\|\nabla\mathcal{F}\|$ can be estimated as:

$$\|\nabla\mathcal{F}\| \sim \frac{\partial C_2}{\partial\omega} \cdot \frac{\partial\omega}{\partial C_2} \sim \gamma_{\mathrm{eff}}^2 \cdot \frac{C_2^2}{\hbar^2}$$

where γ_{eff} is the effective anharmonicity. Convergence is guaranteed when $\gamma_{\mathrm{eff}} < \gamma_{\max} \approx 5$–$10$, which holds for all materials tested here.

Mixing for strongly anharmonic systems. Near structural phase transitions or for large-γ systems (e.g., near-ferroelectric PbTe), the unmodified iteration can overshoot. Linear mixing with parameter $\beta \in (0, 1)$ stabilizes convergence:

$$C_2^{(n+1),\,\mathrm{mixed}} = \beta\,\mathcal{F}(C_2^{(n)}) + (1-\beta)\,C_2^{(n)}$$

Damped iteration with $\beta = 0.5$ was used for PbTe in this work, reducing the effective Jacobian norm by a factor of β. The fixed-point is unchanged; only convergence speed is affected.

Appendix G: Variational Grounding: SOTC as Deterministic SSCHA Quadrature

The Gibbs-Bogoliubov inequality. The exact Helmholtz free energy $F_{\mathrm{exact}} = -k_B T \ln Z_{\mathrm{exact}}$ with

$Z_{\rm exact} = \mathrm{Tr}(e^{-\hat{H}/k_BT})$ includes all anharmonicity and cannot be evaluated exactly. For any trial harmonic Hamiltonian $\hat{H}_0$ with free energy $F_0 = -k_BT \ln Z_0$, the **Gibbs-Bogoliubov theorem** provides an exact upper bound:

$$\boxed{F_{\rm exact} \leq F_0 + \langle \hat{H} - \hat{H}_0 \rangle_0 \equiv F_{\rm var}}$$

Proof. Write $Z_{\rm exact} = \mathrm{Tr}(e^{-\hat{H}_0/k_BT} . e^{-(\hat{H}-\hat{H}_0)/k_BT}) = Z_0 \langle e^{-(\hat{H}-\hat{H}_0)/k_BT} \rangle_0$. Apply Jensen's inequality with the convex function e^x:

$$\langle e^{-(\hat{H}-\hat{H}_0)/k_BT} \rangle_0 \geq e^{-\langle \hat{H}-\hat{H}_0 \rangle_0 / k_BT}$$

Therefore $Z_{\rm exact} \geq Z_0\, e^{-\langle \hat{H}-\hat{H}_0 \rangle_0/k_BT}$. Taking $-k_BT \ln(\cdot)$ (reversing the inequality) gives $F_{\rm exact} \leq F_0 + \langle \hat{H} - \hat{H}_0 \rangle_0$. □

SSCHA as a variational problem. The SSCHA [7, 8] chooses $\hat{H}_0$ as a harmonic Hamiltonian with renormalized IFCs $\mathbf{\Phi}^*(T)$ and minimizes $F_{\rm var}[\mathbf{\Phi}^*]$ with respect to $\mathbf{\Phi}^*$. Setting $\partial F_{\rm var}/\partial \Phi_{ij}^{*\alpha\beta} = 0$ yields the SSCHA self-consistency condition:

$$\Phi_{ij}^{*\alpha\beta} = \left\langle \frac{\partial^2 V_{\rm exact}}{\partial u_i^\alpha \partial u_j^\beta} \right\rangle_{\mathcal{D}[\mathbf{C}^*]}$$

where the average $\langle\cdot\rangle_{\mathcal{D}}$ is over the Gaussian displacement distribution with covariance $\mathbf{C}^*(T)$ determined self-consistently by $\mathbf{\Phi}^*$. This integral is evaluated by SSCHA stochastically: $M = 50$–500 configurations are drawn from $\mathcal{D}[\mathbf{C}^*]$ and the average approximated as a Monte Carlo sum, converging as $1/\sqrt{M}$.

SOTC as deterministic quadrature. The SOTC moment-matching condition (App. C):

$$\frac{1}{K}\sum_{k=1}^{K} u_i^{\alpha,(k)} u_j^{\beta,(k)} = C_{ij}^{\alpha\beta}(T)$$

is precisely the *quadrature node condition* for a K-point **Gauss-Hermite quadrature rule** integrating over the Gaussian $\mathcal{D}[\mathbf{C}]$. The nodes $\{\mathbf{u}^{(k)}\}$ are the SOTC displacement configurations; the SOTC objective $\mathcal{Q}$ is minimized to find them. This provides SOTC with:

1. **A rigorous variational upper bound** $F_{\rm SOTC} \geq F_{\rm exact}$, inherited from the Gibbs-Bogoliubov inequality.
2. **Exponential convergence** in the number of quadrature points K, rather than the $1/\sqrt{K}$ convergence of Monte Carlo; this is the mathematical origin of the dramatic efficiency improvement over SSCHA.
3. **Deterministic reproducibility**: unlike SSCHA, SOTC calculations with the same inputs always produce the same result.

The SOTC self-consistency loop (App. F) simultaneously optimizes the trial Hamiltonian $\hat{H}_0[\mathbf{\Phi}^*(T)]$, placing SOTC rigorously within the variational framework.

Appendix H: Harmonic Free Energy on the SOTC Cell

After self-consistency convergence, the renormalized frequencies $\omega^*_{\mathbf{q}s}(T)$ define the free energy of the best harmonic approximation. The single-mode partition function is:

$$\begin{aligned} Z_\omega &= \sum_{n=0}^{\infty} e^{-\hbar\omega(n+1/2)/k_BT} \\ &= e^{-\hbar\omega/2k_BT} \cdot \frac{1}{1 - e^{-\hbar\omega/k_BT}} \\ &= \frac{1}{2\sinh(\hbar\omega/2k_BT)} \end{aligned}$$

using the geometric series. The single-mode free energy $F_\omega = -k_BT \ln Z_\omega = k_BT \ln(2\sinh(\hbar\omega/2k_BT))$. Summing over all $3N_{\rm SC}$ independent modes:

$$\boxed{F_{\rm harm}^{\rm SOTC}(T) = k_BT \sum_{\mathbf{q},s} \ln\left(2\sinh\frac{\hbar\omega^*_{\mathbf{q}s}}{2k_BT}\right)}$$

Thermodynamic quantities. Differentiating:

$$S_{\rm harm}(T) = k_B \sum_{\mathbf{q},s} \left[\frac{\hbar\omega^*_{\mathbf{q}s}}{2k_BT}\coth\frac{\hbar\omega^*_{\mathbf{q}s}}{2k_BT} - \ln\left(2\sinh\frac{\hbar\omega^*_{\mathbf{q}s}}{2k_BT}\right)\right]$$

$$C_V^{\rm harm}(T) = k_B \sum_{\mathbf{q},s} \left(\frac{\hbar\omega^*_{\mathbf{q}s}}{2k_BT}\right)^2 \frac{1}{\sinh^2\left(\hbar\omega^*_{\mathbf{q}s}/2k_BT\right)}$$

$$\mathrm{ZPE} = F_{\rm harm}(0) = \sum_{\mathbf{q},s} \frac{\hbar\omega^*_{\mathbf{q}s}}{2}$$

$$\mathrm{MSD}(T) = \frac{\hbar}{2M}\sum_{\mathbf{q},s} \frac{|e_i^\alpha(\mathbf{q}s)|^2}{\omega^*_{\mathbf{q}s}} \coth\frac{\hbar\omega^*_{\mathbf{q}s}}{2k_BT}$$

Limiting behavior. At $T \to 0$: $2\sinh(x) \to e^x$, so $F \to \sum \hbar\omega/2$ (ZPE). At $k_BT \gg \hbar\omega$: $\sinh(x) \to x$, giving $F \to k_BT \sum \ln(\hbar\omega_{\mathbf{q}s}/k_BT)$ (classical Dulong-Petit). The heat capacity $C_V \to 3N_{\rm SC}k_B$ at high T, correctly recovering the Dulong-Petit law.

Appendix I: Anharmonic Free Energy: Wick's Theorem and Perturbation Theory

Setup. The full potential $V = V_2 + V_3 + V_4 + \cdots$ includes cubic (V_3) and quartic (V_4) anharmonic terms. In the variational framework (App. G), the first-order anharmonic correction to F_{var} is:

$$\Delta F^{(1)}_{\text{anharm}} = \langle V_3 \rangle_{\mathcal{D}} + \langle V_4 \rangle_{\mathcal{D}}$$

where $\langle \cdot \rangle_{\mathcal{D}}$ is the Gaussian average over displacements with covariance $\mathbf{C}^*(T)$.

Wick's theorem. For the multivariate Gaussian $\mathcal{D}[\mathbf{u}] \propto e^{-\mathbf{u}^T (\mathbf{C}^*)^{-1} \mathbf{u}/2}$, the moment-generating function is:

$$Z[\mathbf{J}] = \langle e^{\mathbf{J}\cdot\mathbf{u}} \rangle_{\mathcal{D}} = \exp\left(\frac{1}{2} \sum_{i,j} J_i C^*_{ij} J_j \right)$$

Proof: Complete the square: $-\frac{1}{2}\mathbf{u}^T(\mathbf{C}^*)^{-1}\mathbf{u} + \mathbf{J}\cdot\mathbf{u} = -\frac{1}{2}(\mathbf{u} - \mathbf{C}^*\mathbf{J})^T(\mathbf{C}^*)^{-1}(\mathbf{u} - \mathbf{C}^*\mathbf{J}) + \frac{1}{2}\mathbf{J}^T\mathbf{C}^*\mathbf{J}$. Shifting $\mathbf{v} = \mathbf{u} - \mathbf{C}^*\mathbf{J}$ leaves a pure Gaussian in $\mathbf{v}$ (integral $= Z_0$) multiplied by $e^{\frac{1}{2}\mathbf{J}^T\mathbf{C}^*\mathbf{J}}$, yielding $Z[\mathbf{J}] = e^{\frac{1}{2}\mathbf{J}^T\mathbf{C}^*\mathbf{J}}$.

Moments follow by differentiation. The $2n$-point correlator is given by the **Wick-Isserlis theorem**:

$$\langle u_{i_1} u_{i_2} \cdots u_{i_{2n}} \rangle_{\mathcal{D}} = \sum_{\sigma \in \mathcal{P}_{2n}} \prod_{\text{pairs } (a,b)\in\sigma} C^*_{i_a i_b}$$

summing over all $(2n-1)!! = (2n-1)(2n-3)\cdots 3 \cdot 1$ distinct pairings. Odd moments vanish identically ($\langle u_{i_1} \cdots u_{i_{2n+1}} \rangle_{\mathcal{D}} = 0$) because the Gaussian is symmetric under $\mathbf{u} \to -\mathbf{u}$.

Cubic term vanishes. Since $\langle V_3 \rangle_{\mathcal{D}} = \frac{1}{6}\sum_{ijk\alpha\beta\gamma} \Psi^{\alpha\beta\gamma}_{ijk} \langle u^\alpha_i u^\beta_j u^\gamma_k \rangle_{\mathcal{D}} = 0$ by the odd-moment rule, the cubic anharmonicity makes **no first-order contribution** to the variational free energy. This is an exact result, not an approximation.

Quartic tadpole contribution. The 4-point Wick expansion gives three pairings. Exploiting the full permutation symmetry of $\Xi^{\alpha\beta\gamma\delta}_{ijkl}$ (simultaneously exchanging any pair of compound indices $(i\alpha) \leftrightarrow (j\beta)$ etc.), all three pairings contribute equally, giving:

$$\boxed{\Delta F^{var}_4 = \langle V_4 \rangle_{\mathcal{D}} = \frac{1}{8} \sum_{ijkl} \sum_{\alpha\beta\gamma\delta} \Xi^{\alpha\beta\gamma\delta}_{ijkl}\, C^{*\alpha\beta}_{ij}(T)\, C^{*\gamma\delta}_{kl}(T)}$$

In the Feynman-diagram representation, this is the **tadpole diagram**: a quartic vertex Ξ_{ijkl} with two propagator loops C^*_{ij} and C^*_{kl} attached. At high temperature $C^*_{ij} \propto k_B T$, so $\Delta F^{var}_4 \propto (k_B T)^2$; quartic anharmonicity produces a super-linear free energy correction that stiffens the phonons with increasing temperature (phonon hardening).

Three-phonon scattering rate. From the 3IFC ensemble (Sec. V.C), phonon lifetimes follow from the Fermi Golden Rule:

$$\frac{1}{\tau_{\mathbf{q}s}(T)} = \frac{\pi\hbar}{8N} \sum_{\mathbf{q}'s',\mathbf{q}''s''} \left|V_3(\mathbf{q}s;\mathbf{q}'s';\mathbf{q}''s'')\right|^2 \Delta_{\mathbf{q}+\mathbf{q}'+\mathbf{q}'',\mathbf{G}}\, f(\omega,\omega',\omega'',T)$$

where V_3 is the three-phonon matrix element (quadratic in polarization vectors, linear in 3IFCs), Δ enforces momentum conservation (including Umklapp processes with reciprocal vector $\mathbf{G}$), and f encodes Bose-Einstein factors for energy conservation. The lattice thermal conductivity follows from the Boltzmann transport equation (relaxation-time approximation):

$$\kappa_L = \frac{1}{NV} \sum_{\mathbf{q},s} C_{\mathbf{q}s}(T)\, v^2_{\mathbf{q}s}\, \tau_{\mathbf{q}s}(T)$$

where $C_{\mathbf{q}s} = \hbar\omega_{\mathbf{q}s}\, \partial n_B/\partial T$ is the modal heat capacity and $v_{\mathbf{q}s} = |\nabla_{\mathbf{q}}\omega_{\mathbf{q}s}|$ is the group velocity. Extraction of 3IFCs from a small SOTC ensemble (5–10 configurations with varied displacement patterns, chosen to independently sample all triplet correlators) is a natural extension of the present framework.

Appendix J: Quasi-Harmonic Approximation: Thermodynamic Identities, EOS Fitting, and Interpolation Accuracy

This appendix provides derivations supporting Sec. II.F.

J.1 Thermodynamic identity for $C_P - C_V$.

Starting from the fundamental relation between the Helmholtz free energy $G(T,V)$ and the thermodynamic response functions, we derive the standard identity used in Eq. (17). Define the pressure $P = -(\partial G/\partial V)_T$ and the isothermal bulk modulus $B_T = -V(\partial P/\partial V)_T = V(\partial^2 G/\partial V^2)_T$. The isobaric heat capacity follows from the Gibbs free energy $\mathcal{G}(T,P) = G(T,V^*) + PV^*$:

$$C_P = -T \left(\frac{\partial^2 \mathcal{G}}{\partial T^2} \right)_P$$

Using the chain rule to relate C_P to $C_V = -T(\partial^2 G/\partial T^2)_V$ at constant volume, one obtains (see, e.g., Ref. [43]):

$$C_P - C_V = -T\, \frac{\left[(\partial^2 G/\partial T\, \partial V)\right]^2}{(\partial^2 G/\partial V^2)_T} = T\, V^*\, \alpha^2\, B_T$$

where the thermal expansion coefficient $\alpha = (1/V^*)(\partial V^*/\partial T)_P$ and the mixed derivative $\partial^2 G/\partial T\, \partial V = -\partial P/\partial T = \alpha B_T$ (the Maxwell relation from the Maxwell relation $(\partial S/\partial V)_T = (\partial P/\partial T)_V$).

This identity is exact within the quasi-harmonic approximation and is the equation used in Sec. II.F.

J.2 Parabolic equation of state and static bulk modulus.

For three volumes $V_- < V_0 < V_+$ with equal spacing $\Delta V = V_+ - V_0 = V_0 - V_-$, the static energy $E_0(V)$ is fitted by a second-degree polynomial (exact through the three points):

$$E_0(V) = a + b(V - V_0) + c(V - V_0)^2$$

The fit coefficients are determined by the system:

$$c = \frac{E_0(V_+) - 2E_0(V_0) + E_0(V_-)}{2(\Delta V)^2}, \quad b = \frac{E_0(V_+) - E_0(V_-)}{2\Delta V}$$

At equilibrium ($dE_0/dV = b + 2c(V_{\rm eq} - V_0) = 0$), the zero-pressure equilibrium volume is $V_{\rm eq} = V_0 - b/(2c)$. The static bulk modulus is:

$$B_0 = V_0 \left.\frac{d^2E_0}{dV^2}\right|_{V_0} = 2c\,V_0$$

in units of eV/Å^3 (convert to GPa via 1 eV/Å^3 = 160.218 GPa). For a volume step $\delta = \Delta V/V_0 = 0.02$ and typical Al parameters ($V_0 = 16.48$ Å^3, $c \approx 0.51$ eV/Å^6), this gives $B_0 \approx 77$ GPa, in good agreement with the experimental zero-temperature value of 76–79 GPa.

The parabolic EOS is equivalent to the second-order Birch-Murnaghan EOS when the volume range is small ($\delta \leq 0.05$). For larger volume perturbations, the third-order Birch-Murnaghan form $E_0(V) = E_0 + \frac{9V_0B_0}{16}\left\{\left[(V_0/V)^{2/3} - 1\right]^3 B_0' + \left[(V_0/V)^{2/3} - 1\right]^2\left[6 - 4(V_0/V)^{2/3}\right]\right\}$ should be used. For $\delta \leq 0.02$ the difference between parabolic and third-order Birch-Murnaghan is < 0.01 meV/f.u. in E_0 over the interpolation range, justifying the parabolic approximation.

J.3 Accuracy of the quadratic interpolation of $F_{\rm vib}(T,V)$.

The vibrational free energy $F_{\rm vib}(T,V)$ is interpolated quadratically in V from the three SOTC phonon calculations at $V_\pm, V_0$:

$$F_{\rm vib}(T,V) \approx a_F(T) + b_F(T)(V - V_0) + c_F(T)(V - V_0)^2$$

The error in this approximation arises from the fourth-order term $\frac{1}{24}F^{(4)}_{\rm vib}(T,V_0)(\Delta V)^4$. Using the mode Grüneisen expansion $\omega_{\mathbf{q}s}(V) = \omega^{(0)}_{\mathbf{q}s}\exp(-\gamma_{\mathbf{q}s}\ln(V/V_0))$, the volume derivatives of $F_{\rm vib}$ are:

$$\frac{\partial F_{\rm vib}}{\partial V} = -\frac{1}{V}\sum_{\mathbf{q}s}\gamma_{\mathbf{q}s}\hbar\omega_{\mathbf{q}s}\left[n_B(\omega_{\mathbf{q}s}) + \tfrac{1}{2}\right]$$

where $n_B(\omega) = (e^{\hbar\omega/k_BT} - 1)^{-1}$ is the Bose-Einstein factor. The fourth derivative scales as $\gamma^4\langle\omega^4\rangle_T/V^4$, giving a fractional interpolation error in $F_{\rm vib}$:

$$\frac{\left|\delta F^{(4)}_{\rm vib}\right|}{F_{\rm vib}} \sim \frac{\gamma^4\delta^4}{4!} \approx \frac{(2.0)^4(0.02)^4}{24} \approx 3\times10^{-7}$$

using a representative Grüneisen parameter $\gamma = 2$ (upper bound for PbTe) and $\delta = 0.02$. This is well below the meV/f.u. accuracy threshold for thermodynamic properties, validating the quadratic interpolation for $\delta \leq 0.02$.

J.4 IFC extrapolation $\Phi(V_+) = 2\Phi(V_0) - \Phi(V_-)$.

When only two SOTC runs are available (at V_0 and V_-), the IFCs at the expanded volume V_+ can be estimated by linear extrapolation. To first order in $\delta = \Delta V/V_0$, the volume dependence of the IFCs is $\Phi_{ij}(V) = \Phi_{ij}(V_0)[1 - \gamma^\Phi_{ij}\ln(V/V_0)]$, where γ^Φ_{ij} is the bond Grüneisen parameter. The linear extrapolation is then:

$$\Phi(V_+) = 2\Phi(V_0) - \Phi(V_-) + \mathcal{O}(\delta^2)$$

The error is second-order in δ: $|\delta\Phi/\Phi| \sim \gamma^{\Phi 2}\delta^2/2$. For $\gamma^\Phi \leq 2$ and $\delta = 0.02$, the fractional IFC error is $\leq 0.04\%$, which propagates to a $< 0.1\%$ error in phonon frequencies and < 0.1 J/(mol·K) error in C_V. The extrapolation is therefore accurate to within the convergence tolerance of the SOTC self-consistency loop.

J.5 Mode Grüneisen parameters and $\alpha(T)$ from the Grüneisen shortcut.

The mode Grüneisen parameter (Eq. 18 of the main text) can be rewritten in terms of the logarithmic frequency-volume derivative:

$$\gamma_{\mathbf{q}s} = -\frac{d\ln\omega_{\mathbf{q}s}}{d\ln V}$$

The macroscopic Grüneisen parameter $\gamma_{\rm eff}(T)$ is defined as the C_V-weighted average:

$$\gamma_{\rm eff}(T) = \frac{\sum_{\mathbf{q}s} c_{\mathbf{q}s}(T)\,\gamma_{\mathbf{q}s}}{\sum_{\mathbf{q}s} c_{\mathbf{q}s}(T)}$$

where $c_{\mathbf{q}s}(T) = k_B(\hbar\omega_{\mathbf{q}s}/2k_BT)^2\sinh^{-2}(\hbar\omega_{\mathbf{q}s}/2k_BT)$ is the modal heat capacity. To derive $\alpha(T) = \gamma_{\rm eff}C_V/(3B_0V_0)$, begin from the equilibrium condition $dG/dV = 0$ at $P = 0$:

$$\frac{dE_0}{dV} + \frac{\partial F_{\rm vib}}{\partial V} = 0$$

Differentiating with respect to T and using $\alpha = (1/V^*)(dV^*/dT)$:

$$\frac{d^2E_0}{dV^2}\frac{dV^*}{dT} + \frac{\partial^2 F_{\rm vib}}{\partial V\,\partial T} = 0$$

$$\Rightarrow \quad \alpha = \frac{1}{V^*}\frac{dV^*}{dT} = -\frac{1}{V^*}\frac{\partial^2 F_{\rm vib}/\partial V\,\partial T}{d^2E_0/dV^2}$$

The numerator evaluates to $\partial^2 F_{\rm vib}/\partial V\,\partial T = -(\partial/\partial T)\sum_{\mathbf{q}s}\gamma_{\mathbf{q}s}\hbar\omega_{\mathbf{q}s}(n_B+\frac{1}{2})/V = -\gamma_{\rm eff}(T)C_V(T)/V_0$. The denominator is $d^2E_0/dV^2 = B_0/V_0$. Substituting:

$$\alpha(T) = \frac{\gamma_{\rm eff}(T)\,C_V(T)}{V_0\cdot B_0/V_0\cdot V_0} = \frac{\gamma_{\rm eff}(T)\,C_V(T)}{B_0\,V_0}$$

The factor of 3 difference from the isotropic approximation ($\alpha_{\rm vol} = 3\alpha_{\rm lin}$) appears when C_V is expressed per unit cell rather than per volume. For a cubic crystal with volumetric α and C_V per formula unit in J/(mol·K), the formula becomes $\alpha = \gamma_{\rm eff}C_V/(3B_0V_0N_A)$, matching Eq. (19) of the main text. This derivation confirms that the Grüneisen shortcut is exact in the quasi-harmonic limit when higher-order volume derivatives of $F_{\rm vib}$ are neglected (valid for $\delta \le 0.02$, see J.3).

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